

AKADEMIKA

FCL-04

**FIBER OPTIC ADVANCE DIGITAL
COMMUNICATION KIT**

**EXPERIMENTAL & SERVICE
MANUAL**

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. Basic electronics is one of the important subject need to teach while learning electronics. It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook as they give important points on safety during the installation, use and maintenance. Keep this handbook for any further help.

After unpacking, set all accessories in order so that they are not lost and check the kit integrity. In particular, check that the kit is integral and shows no visible damage.

Before connecting the power supply to the kit, be sure that the jumpers and connecting chords are connected correctly as per experiment.

In **FIBER OPTIC COMMUNICATION LAB** modules, signals are to be monitored with an oscilloscope as explained in the manual. The scope input channel should be ac coupled, unless otherwise indicated. For observation of signals on oscilloscope either use X10 (Attenuation Probe) or 180 Ω resistance in series with normal oscilloscope probe.

This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of any fault or bad operation, turn off the power supply and do not tamper the kit. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

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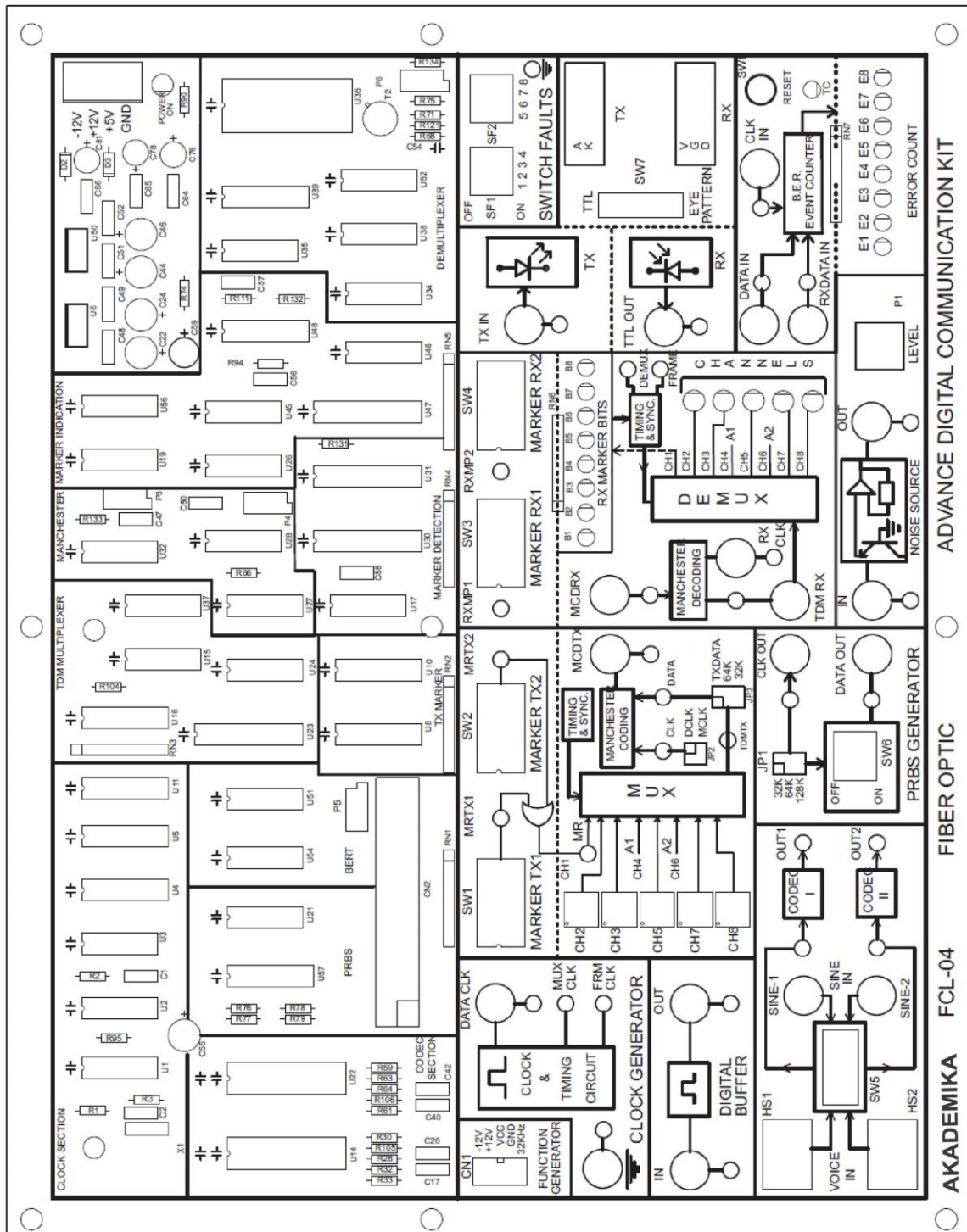
TECHNICAL SPECIFICATIONS

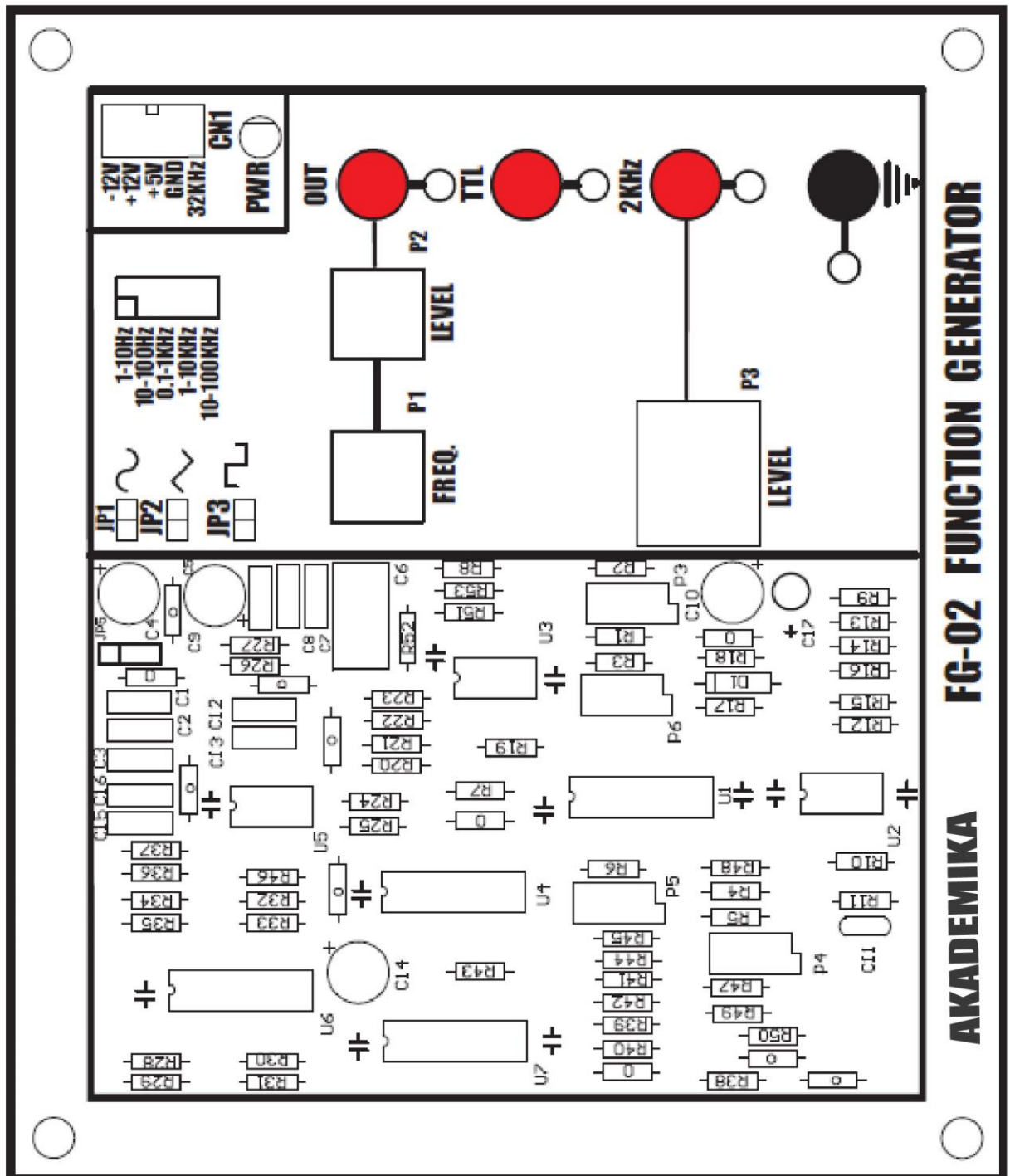
Transmitter	: 1 No. Siemens Fiber Optic LED Peak wavelength of emission 950 nm (invisible) infrared LED (SFH450V)
Receiver	: 1 No. Siemens Photo Detector with TTL logic output (SFH551V)
Multiplexing	: Time Division Multiplexing & Demultiplexing
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Mux / Dmux Rate	: 64 Kbps
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Acceptance Angle	: 60 degrees
Fiber Diameter	: 1000 microns
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FUNCTIONAL BLOCK FCL-04



FG-02

INTRODUCTION

FIBER OPTICS TECHNOLOGY is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T, sprint have saturated the airways & print media with advertisements heralding this bright & new technology, Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

And their enthusiasm is valid. Today medical fiber optic systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologies ever come close to the fantastic growth rate that market analyst love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes, workplaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronic newspapers become feasible. Two-way fiber optic will allow you to “attend college” in the comfort of your home.

Anyone who takes an interest in fiber optic & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few “Experts” exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons exciting with the prospect of making new discoveries, finding new applications for the technology.

This kit is an introduction to fiber optics communication technology for instructors, students & hobbyist. Welcome to the fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & - yes – even fun.

ABOUT FIBER OPTICS

Before fiber optics came along, the primary means of real time data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic (radio) waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances – using light energy. Although initially a very controversial technology fiber optics has today been shown to be very reliable & cost effective. Light as utilized for communication has a major advantages because it can be manipulated (modulated) at significant higher frequencies than electrical signals can. For example, a fiber optic cable can carry up to 100 million times more information than telephone line! The fiber optic cable has lower energy loss & wider bandwidth capabilities than copper wire.

As you will learn, fiber optic communication is a quite simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today. Fiber optics became reality when several technologies came together at once. It was not an immediate process, nor was it easy, but it was most impressive when it occurred.

An example of one critical product, which emerged from that technological merger, was the semiconductor LED that we used in our kit.

How Optical Fiber Works?

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Air & water are a case in point as shown in fig. 1. The light ray traveling through air actually is bent as it enters the water. The amount of bending depends on the refractive indices of the two materials involved & also on the angle of the incoming (incident) ray of light as it strikes the interface. The angle of the incident ray is measured from a line drawn perpendicular to the surface. The same is true for the angle of the outgoing (transmitted) ray of light after it has been bent.

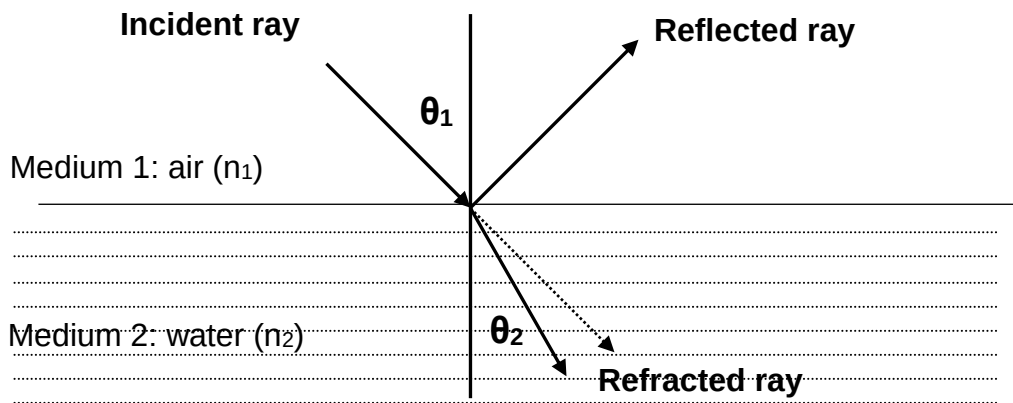


FIG. 1: The different portions of a light ray at a material interface.

The mathematical relationship between the incident ray & the refracted ray is explained by **Snell's Law**

$$n_1 \cdot \sin \theta_1 = n_2 \cdot \sin \theta_2$$

in which n_1 & n_2 are the refractive indices of the initial and secondary materials respectively, and θ_1 and θ_2 are the incident and transmitted angles.

Snell's Law says that refraction (bending) of light cannot take place when the angle of incidence grows too large (as when light travels from a material with a high refractive index to one with a low refractive index). If the angle of incidence exceeds a certain critical value (in which the product of n_1 & the sine of the angle equals or exceeds one.) light cannot exit.

If light cannot exit the material, it is reflected. The angle that is reflected is equal to the angle of incidence. The phenomenon just described is called Total Internal Reflection & it is what keeps light inside an optical fiber.

Types of Optical Fiber

The simplest fiber optic cable consists of two concentric layers of transparent materials. The inner portion (the core) transports the light. The outer covering (the cladding) must have a lower refractive index than the core, so the two are made of different materials. The cable used in this kit also has a jacket to protect the optical properties of the core and cladding.

Optical fiber is generally made from either plastic or glass. This kit uses plastic fiber, which is very easy to terminate, & does not require special tools. Plastic fiber is generally limited to uses involving distance of less than 100 meters because of high attenuation (higher loss).

Glass fiber on other hand has very low attenuation (low loss) & hard to cut, requires special end connections & is more expensive.

The core of fiber is typically made of silica doped with impurities, which increases the refractive index relative to pure silica. The cladding, which surrounds the fiber core & creates an optical interface, is typically made from pure silica. The outer buffer coating is a plastic cover.

The first type of fiber optic cable put to use was called Step Index. In this design the cladding has a different index of refraction from the core. The light bounces off the sides & is reflected back into the fiber core. The problem with this design is that the reflected light must travel a slightly longer distances than that which travels down the center of the fiber, thus limiting the max. Transmission rate. This design was improved with the use of Graded Index Fiber. In this design, the index of refraction decreases in proportion to the distance away from the center of the fiber core. The light moves more quickly in the outer portion, thus compensating for the additional distance. The change in index also has the effect of 'bending'. The light reflects back towards the core. This change increases the transmission capacity. In the newest Single Mode design, the diameter of the fiber core is made so small that all the light travels in a straight line. Even the latest fiber optic facility in use today uses less than 5% of the max. Theoretical capacity of a Single Mode Fiber.

Single Mode V/S Multimode

There are essentially two different types of fiber optic transmission schemes in use. Multimode fiber optic systems are typically used in short haul applications such as LANS & FDDI. Recent growth in multimode installations has been driven by LAN/FDDI applications. The term Multimode means that the diameter of the fiber optic core is large enough to propagate more than one mode (electromagnetic wave). Because of the multimode modes, the pulse that is transmitted down the fiber tends to become stretched over distance. This is referred to as Modal Dispersion & has the effect of reducing available bandwidth.



FIG. 2: Modal dispersion of light

Typical wavelengths used in multimode applications are 850 nm & 1300 nm. There are other wavelengths used (e.g. 660 nm for plastic fiber, 820/870 nm) in specialized applications such as the military. The vast majority of multimode applications are 850 or 1300 nm wavelengths. Most multimode cables in use

today are graded index in type with a cladding diameter of 125 microns & fiber core diameter of 62.5 or 50 microns (Typically referred to as 62.5/125 or 50/125 fiber)

Single Mode Systems are generally used in long haul applications (10's or 100's of kilometers) & local loop applications in both the carrier & CATV markets. Some large utilities such as electrical power companies also utilize single mode fiber optic backbones. As the name implies single mode fiber is designed to propagate only one mode of light. It is therefore not affected by modal dispersion & has higher bandwidth capacity. It is considered to be higher quality transmission system. Wavelengths used in single mode applications are 1300 & 1550 nm. The majority of today's installed systems uses 1300 nm wavelength technology & is designed for higher speed applications. These systems are more sensitive to back reflections from connectors & sharp cable bends. They often require additional tests, such as Optical Return Loss, to be run. Most single mode cables in use today are step index in type with cladding diameter of 125 micron & fiber core diameter of 9 micron (Typically referred to as 9/125 fiber).

Advantages of Fiber Optics over Conventional Copper Cables

- Much greater bandwidth (Several orders of magnitude are theoretically possible).
- Immunity to electrical disturbances, ground loops, cross talks etc. in addition, no EMI radiation is generated)
- Much lighter. A 3-inch bundle of 900 twisted copper pairs can carry about 21,000 channels of traffic & weighs about 25,000 pounds/mile. A ½ inch fiber link with 12 fiber strands can carry about 3,00,000 channels & weighs about 200 pounds/mile. In addition less duct space is required to make the cable within building.
- Better in hostile environment, not affected as much by temperature, water etc.
- Lower transmission loss. Typical loss on a fiber link is 0.2 dB/Km. On a copper based facility, one can usually expect at least 5 db/km. For longer links, fewer repeaters are required.
- Better security as it is not possible to simply bridge onto the facility & monitor the traffic.

With all these properties of optical fiber, it is becoming very important in sensing field also. Most of the properties of optical fiber are useful for sensors. Further optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

ABOUT FCL-04

The purpose of Fiber Optic Advance Digital Communication Kit is to provide you hands on experience on the various Fiber Optic & Digital Communications Techniques. In preparing this manual, we assumed you have a basic grasp of digital & transistor circuits. The experiments that can be performed with this Lab have been designed in a ladder fashion.

You have to start from the first step & then go for the next. An experiment starts with basic principles of fiber optics & then go on to more sophisticated techniques. Every advanced experiment uses concept, procedures, results of the previous experiments. So you are advised not to skip in between experiments. In this manual before starting the procedure of every experiment, objective of the experiment & basic theory related to that is given. You are advised to read it before starting the experiment because it will help you to understand the experiment very well. Also wherever necessary refer the datasheets of various components provided on the backside of the manual.

Before starting the experiments check following things in the Lab

- Plastic fiber of 1Meter length
- 2 Nos. of telephone handsets
- 1 No. Power Supply
- Link set containing Patch chords, Jumpers & crocodile clips wires
- Experimental and Circuit Description Manuals

Optical Fiber Preparation Instructions

The fibers provided with this Lab are multimode, step index, 1000-micron plastic fiber & are factory prepared for use. They are not require to prepared again even after several operations, if handled carefully. In case the fiber is broken or damaged you have to prepare each end of the optical fiber cable very carefully so it transmits light effectively.

- Cut off the ends of the cable with a single-edge razor blade or sharp knife. Try to obtain a precise 90-degree angle.
- Wet the polishing paper with water or light oil & place it on flat, firm surface. Hold the optical fiber upright, at right angle to the paper & polish the fiber tip with a gentle “figure8” motion. You may get the best result by supporting the upright fiber against some flat object such as a portion of the printed circuit board.
- Using an 18-gauge wire stripper, remove 3 mm of the jacket from the end of the fiber. Be careful not to nick the buffer in the process or you will be losing light later.

CIRCUIT DESCRIPTION

1.1 Clock And Timing

The 2.048MHz crystal oscillators generate a 2MHz clock. U1 (IC 74HCT14) is used around crystal for generation of clock. This clock is further divided by counters U4, U5 & U11 (IC 74HC193) to generate 512KHz, 128KHz, 64KHz, 32KHz and 8KHz frequencies. These are generated for the complete time division multiplexed transmission of the data from 8 channels. The data bits are transmitted at the rate of 512 KHz. Each channel is selected for on period of 1/64 KHz. Thus per channel 8 bits of data is transmitted. The channel selection frequency is 8 KHz. Thus a single frame of 8 channels exists for a period of 125uS. Marker selection clock is also generated here. Data clock is 512 KHz, Multiplexer clock is 64 KHz and frame clock is 8 KHz.

1.2 CODEC 1 & 2

The voice signals from telephone handsets or analog signal from external signal source are processed and applied to CODEC U14 & U22 (MC 145502). The analog to digital conversion process inside the CODEC samples the signal at a frequency of 8 KHz and this sampled digitized data is applied to the multiplexer input.

Present techniques of voice communication use standards such as A-law / μ -law companded PCM voice coding at 64 kbits/sec. When the analog speech signal is converted to pulse code modulation, it is first filtered using a low pass filter with a cut-off at about 3.4 KHz. The analog signal is sampled at 8 KHz as per the Nyquist Criteria. Each sample is quantized and coded into eight bits per sample.

Actually the voice signal features something undesirable and that is its amplitude range varies greatly from one conversation to another. If the quantization levels are uniformly spaced then it certainly creates problems. If the amplitude of the signal is small, quantization levels have to be closely spaced. This gives proper resolution. But if the signal amplitude is large than this fine resolution will result in increasing the number of code bits.

Hence, normally a technique of unequal spacing of quantization levels is used. In the Fiber trainer kit, the audio signal from the telephone handset or analog signal from external signal source is processed using telephone interface chip U13 & U20 (TEA1062A) and amplifier circuit. It is then applied to voice coder. This chip actually performs analog to digital conversion and vice-versa.

The digital data output is in pulse code modulated form. The CODEC chip used exhibits both A-law and μ -law companding techniques. Here, we have selected A-law of companding. The digital output of CODEC is connected as i/p to the multiplexer. The received digital data is again converted into analog form by the chip. The clock considerations and other features of the CODEC chip used in this kit can be studied in details from the datasheets provided with the manual.

1.3 Manchester Coding

Refer to the circuit diagram as shown later in this manual sheet **3 of 11** of **FCL-04**. The data to be coded is applied to a flip flop U27 (IC 74HCT74) by selection jumper JP3. An inverted data clock is given as clock input to this flip-flop by selection jumper JP2. The data output from the flip-flop is thus half bit delayed (delayed by one clock pulse). This delayed data is then EX-ORed U28 (IC 74HC86) with the data clock. Thus we get output in the Manchester coded data form. It is then transmitted through fiber link.

1.4 Manchester Decoding

Refer to the circuit diagram as shown later in this manual sheet **3 of 11** of **FCL-04**. The received data is Manchester decoded to separate clock and data and is applied to a demultiplexer.

The received data is integrated using RC integrator U28 (IC 74HC86). The output of this integrator is EX-ORed U28 (IC 74HC86) with the received data. The output of this EX-OR gate gives a low going narrow pulse for each transition of the received data. This signal is applied to a monostable multivibrator U32 (IC 74LS121), which is configured to give a high pulse for each high to low transition at its input. The on period of this output waveform is adjusted such that the multivibrator output is of 512 KHz. This is the received clock, which is then applied

as clock input to Flip-flop U27 (IC74LS74). The output of this Flip-flop is the decoded data. Note that the output is taken at the inverted output pin of Flip-flop. The received data is one clock period shifted. This is due to the particular coding-decoding circuitry used.

1.5 Marker Generation

The marker setting is achieved using switches SW1 & SW2. The two eight bit markers are alternately transmitted at the start of each frame. U8 & U10 (IC 74LS165) are used respectively for Marker-1 and Marker-2 to convert them from parallel to serial form. These signals are ANDed with Marker selection clock and final outputs are ORed to get one channel data.

1.6 Time Division Multiplexing

In case of communication systems, signals which are transmitted usually carry voice or video information with them & are interpreted by human eye or ears, which has slow response. Persistence of vision as well as of hearing has given rise to the concept of time division multiplexing. In time division multiplexing various signals are sampled & transmitted for a fixed duration of time one after the other. At the receiving end these signals are extracted in the same order & form of transmission.

To implement this scheme we have used 8-channel digital multiplexer at transmission end with clock generator for timing of signals. One channel is reserved for marker transmission, two channels for voice data transmission, five channels take their inputs from five data switches. Each channel has a data rate of 64 Kbits / Sec.

Now the repetition rate of frame depends on channel sampling frequency. Since we are transmitting audio signals on these channels, we should sample channel at least twice the highest frequency component in audio signal, which is 4 KHz. This determines frame frequency of 8 KHz. Within this period of 125 micro second. we transmit 8 channels hence, each channel ON period comes about $125/8$ micro second. This corresponds to the frequency of 64 KHz. lastly, since we transmit 8 bits of data per channel, data rate can be derived as $15.625 / 8 = 1.953125$ & $1 / 1.953125 = 512$ KHz.

The different frequencies required for the generation of a frame are obtained from the clock generation circuit. The synchronization between these three frequencies is achieved using a stable crystal clock and the divider circuitry of a counter. The rising edge of frame frequency 8KHz is synchronized with the Marker generation block. The marker is generated by a parallel to serial shift register and it always appears in the first channel of a frame. Channels 2, 3, 5, 7 & 8 in frame receive data from 5 switches. The digitized data from CODEC-I and from CODEC-II appears in 4th and 6th slots respectively.

1.7 Time Division Demultiplexing

The demultiplexer forms the data distribution system from where the samples are separated and are applied to corresponding channels.

U36 (74HC154) is used to generate channel selection signals. U34, & U38 separates out individual channel signals from the TDM received data.

The switch channels directly drive the LED's and the decoded voice data is again applied to the respective CODEC chips for decoding of digital data to analog form.

1.8 Marker Detection

Marker used in Time Division Multiplexing is a unique bit pattern placed at some fixed position in the frame & is used to determine the start of the frame at the receiver. Sometimes a different marker is used in alternate frames to counter the possibility of data bits containing the marker sequence generating a false marker. The marker is supposed to be a unique bit pattern in the data stream, which is identified at the receiver to identify the beginning of frame. Question arises, what happens if the marker pattern is contained elsewhere as data?

Using the two 8-bit markers hundreds of combinations can be formed for TDM communication. Few settings in the marker pattern may match the data and the receiver comparator may generate a false detection pulse, disturbing the TDM communication.

To avoid this situation the markers are set in a way such that the detection pulses are generated at proper time and proper TDM communication is achieved. It is highly impossible that the data will have this pattern repeating at the same position in every frame. However, to avoid even this situation, the marker is sometimes chosen to be different in alternate frames.

Same received data is applied to the marker detection circuitry where low going pulses are generated corresponding to the each marker received and are used as synchronizing signal for timing circuitry. There by two markers are separated from the Multiplexed data. U30 & U31 (IC 74LS688) are used here.

1.9 Marker Indication

From the received markers marker-1 is separated out and given to serial to parallel converter U46 (IC 74HC164). Parallel converted marker is passed through buffer latch U47 (IC 74HC374) and output of latch drive LED's to indicate received marker pattern.

1.10 Fiber Optics Transmitter

Refer to the circuit diagram as shown later in this manual sheet **7** of **11** of **FCL-04**. We have used driver circuitry based on a standard TTL gate to drive a NPN transistor (T2), which modulates the LED SFH450V source (Turns it ON and OFF).

The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver.

Fiber optic transmitters are typically composed of a buffer, driver & an optical source. The buffer section provides both electrical connection & isolation between the optical transmitter & analog information source. The driver section provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern.

FCL-04 kit supply SFH450V LED transmitter. The LED SFH450V operates outside the visible light spectrum. Its optical output is centered at near infrared wavelength of 950nm. This LED is coupled to the transistor driver in common emitter mode. In the absence of input signal there is no output light. When signal is present transistor works in saturation state and it gives bias to transmitter LED, which emits full light. Optical signal is then carried over by the optical fiber.

1.11 Fiber Optics Receiver

There are various methods to configure detectors to extract digital data. Usually detectors are of linear nature. We have used a photodetector having TTL type output (SFH 551/V). Usually it consists of integrated photodiode, transimpedance amplifier and level shifter.

1.12 Digital Buffer

U15 (IC 74HCT14) is used as TTL Buffer

1.13 Bit Error Rate Counter

Transmitted data and received data bits are passed through D-flip-flop U51 (IC 74HCT74). Output of D-flip-flop is then EXORed using U37 (IC 74HC86) and then ANDed with inverted clock using U56 (IC 74HC08). This bit pattern is given as clock input to decade counters U52 (IC 74HC393). Output of counters drive LED's to indicate the error in the received bits in the form of decimal number. Counter is reset using switch SW8.

1.14 PRBS Generator

PRBS generator consists of 4bit dipswitch to change the pattern of PRBS sequence. Shift register U57 (IC 74LS195) is used to generate PRBS.

1.15 Switch Faults

Eight switch faults are provided to insert logical faults in various sections of the circuit.

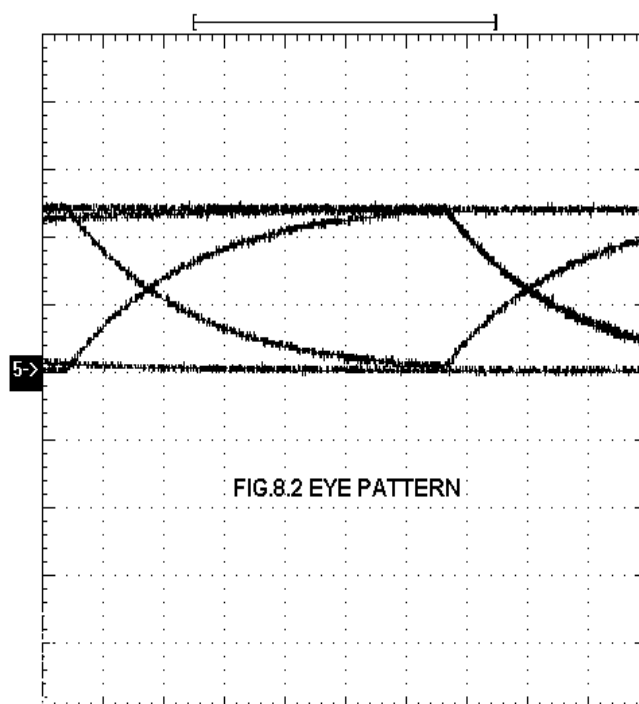
1.16 Noise Source

The noise source consists of a Transistor T1 (BC547) whose base is grounded, collector is left open and emitter is given supply voltage. In this biasing condition it produces a white noise in the order of some tens of mV. The noise is amplified by two amplifier stages U43 (IC TL084) with a potentiometer P5 for level regulation.

SWITCH FAULTS

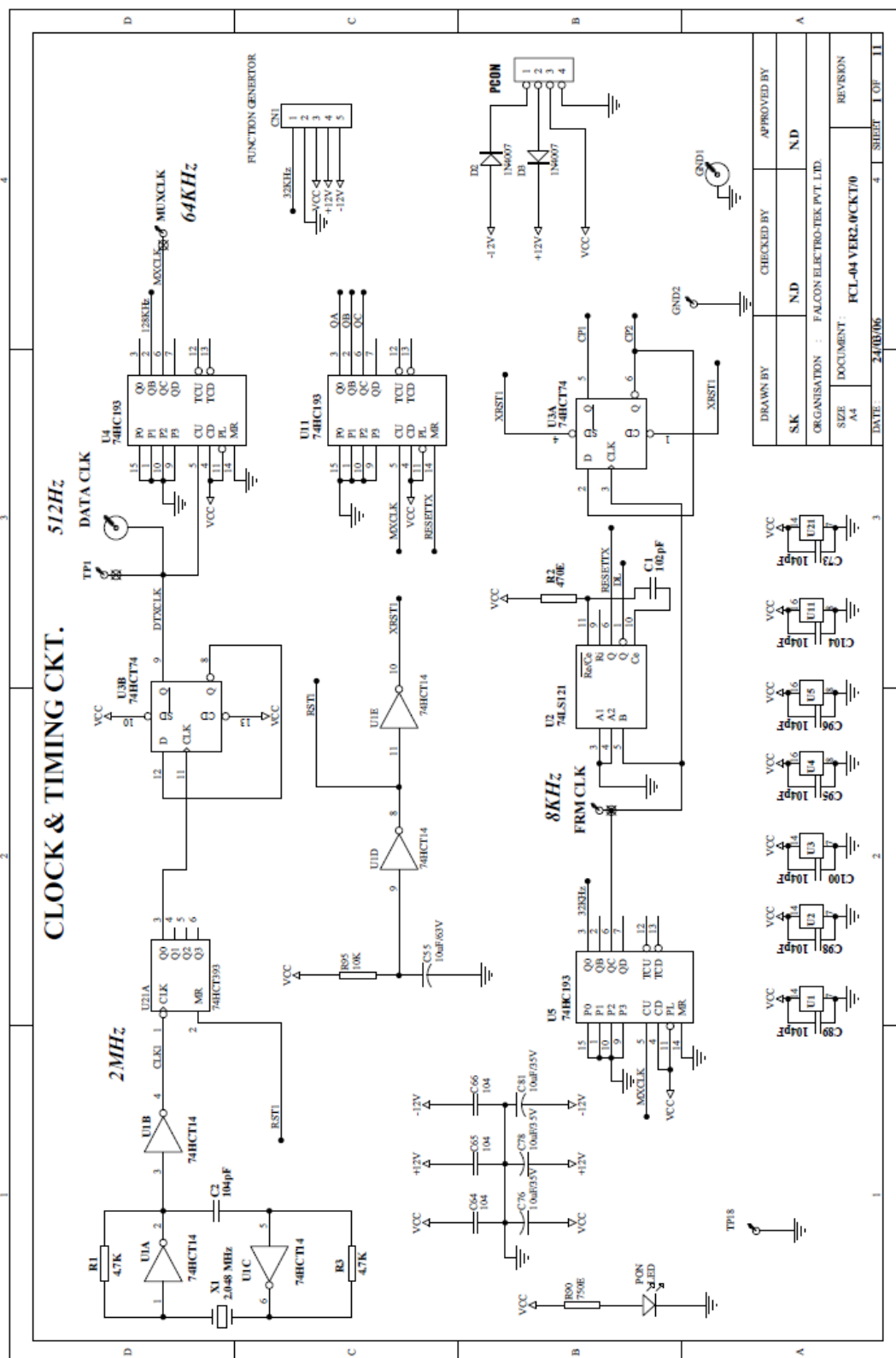
Switch fault section on FCL-04 is provided with 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. In order to keep FCL-04 kit fully operational, all switches should be set to OFF position before use. Switch faults are introduced one at a time.

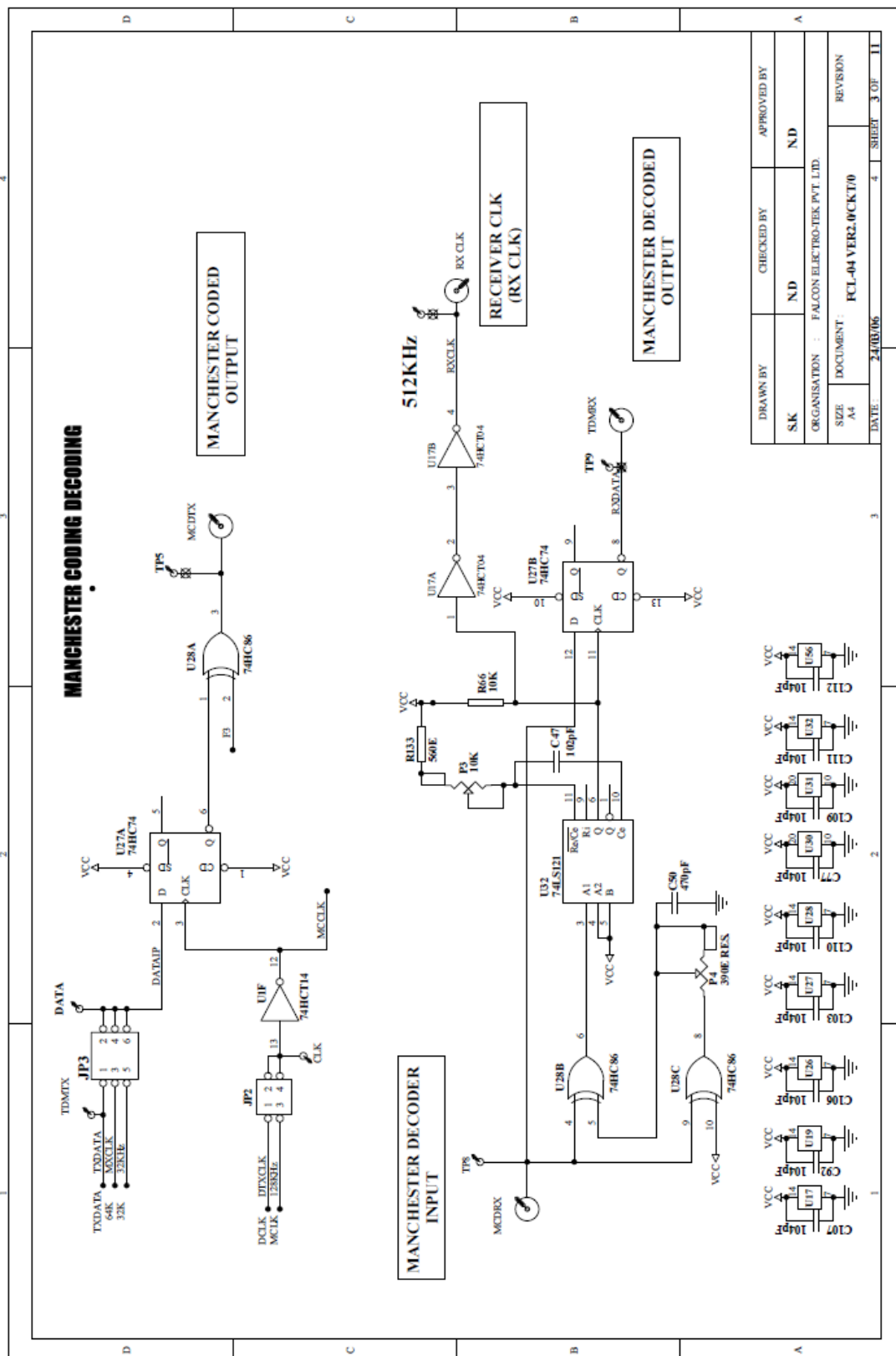
- Put switch **1 (SF1)** in Switch Fault section to ON position. Clock 8KHz of U16 (4051) is removed. Output channels loose synchronization. All channels are affected.
- Put switch **2 (SF1)** in Switch Fault section to ON position. Transmitter markers are removed at the U16 (4051) mux I/P. Hence data is transmitted without marker & hence no sync pulse at receiver and synchronization is lost at Output.
- Put switch **3 (SF1)** in Switch Fault section to ON position. This will open pin 2 of U28a (XOR) in Manchester coding section. MCDTX Output gets disturbed at TP5 and hence Output channels at receiver get disturbed. (No coding takes place & hence MCD clock recovery is disturbed and no demultiplexing takes place.
- Put switch **4 (SF1)** in Switch Fault section to ON position. This will open pin 19 of CODEC2 (U22). MSIRX2 signal of CODEC2 is removed. Output sine wave of CODEC2 gets disturbed.
- Put switch **5 (SF2)** in Switch Fault section to ON position. This will open pin 5 of U36 (74154) demux IC. The channel 5 is not enabled at Output. No LED indication for channel 5.
- Put switch **6 (SF2)** in Switch Fault section to ON position. This will open pin 11 of U45, RXSYN signal is removed at the receiver section & the clock recovery is disturbed, hence the receiver channels are disturbed.

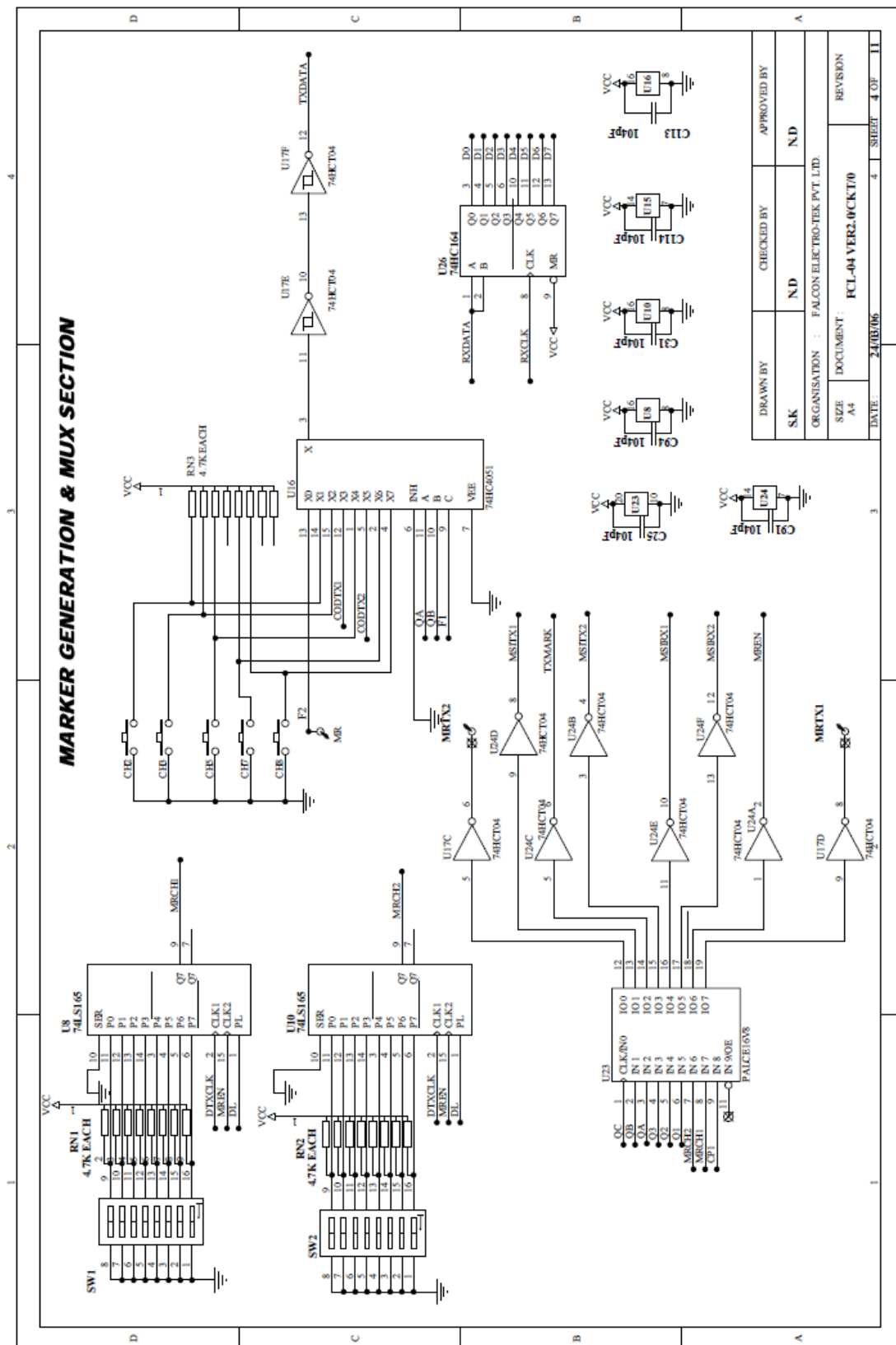


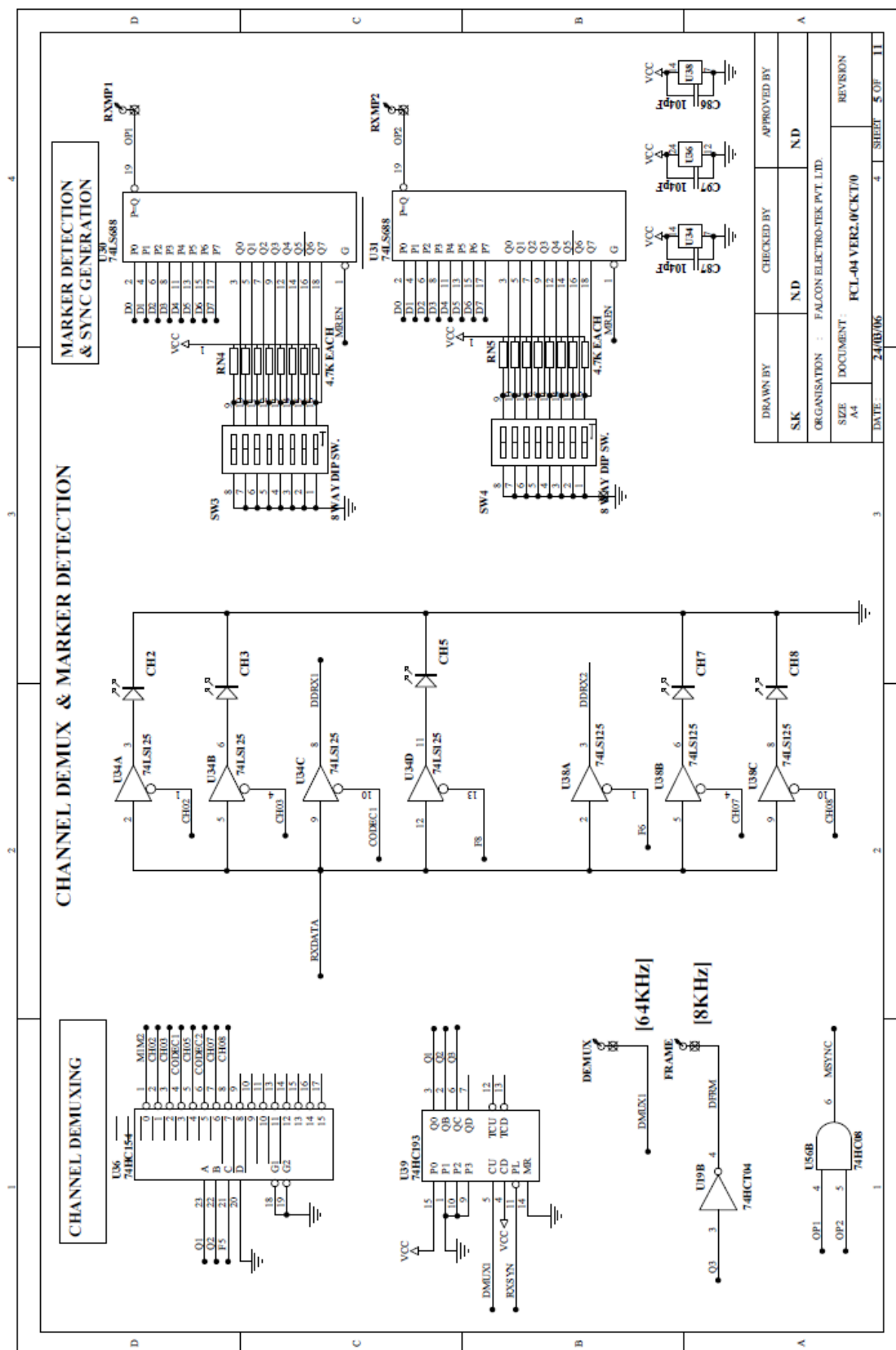
- Put switch **7 (SF2)** in Switch Fault section to ON position. This will open pin 1 of IC U38 (74LS125). The CODEC2 (U22) channel selection is removed and no CODEC response for analog signal or voice channel.
- Put switch **8 (SF2)** in Switch Fault section to ON position. This will open pin 21 of U36 (74154) demux IC. The demux clock Q3 is removed which affects the Output channel demultiplexing, channel indication and bit indications are disturbed.

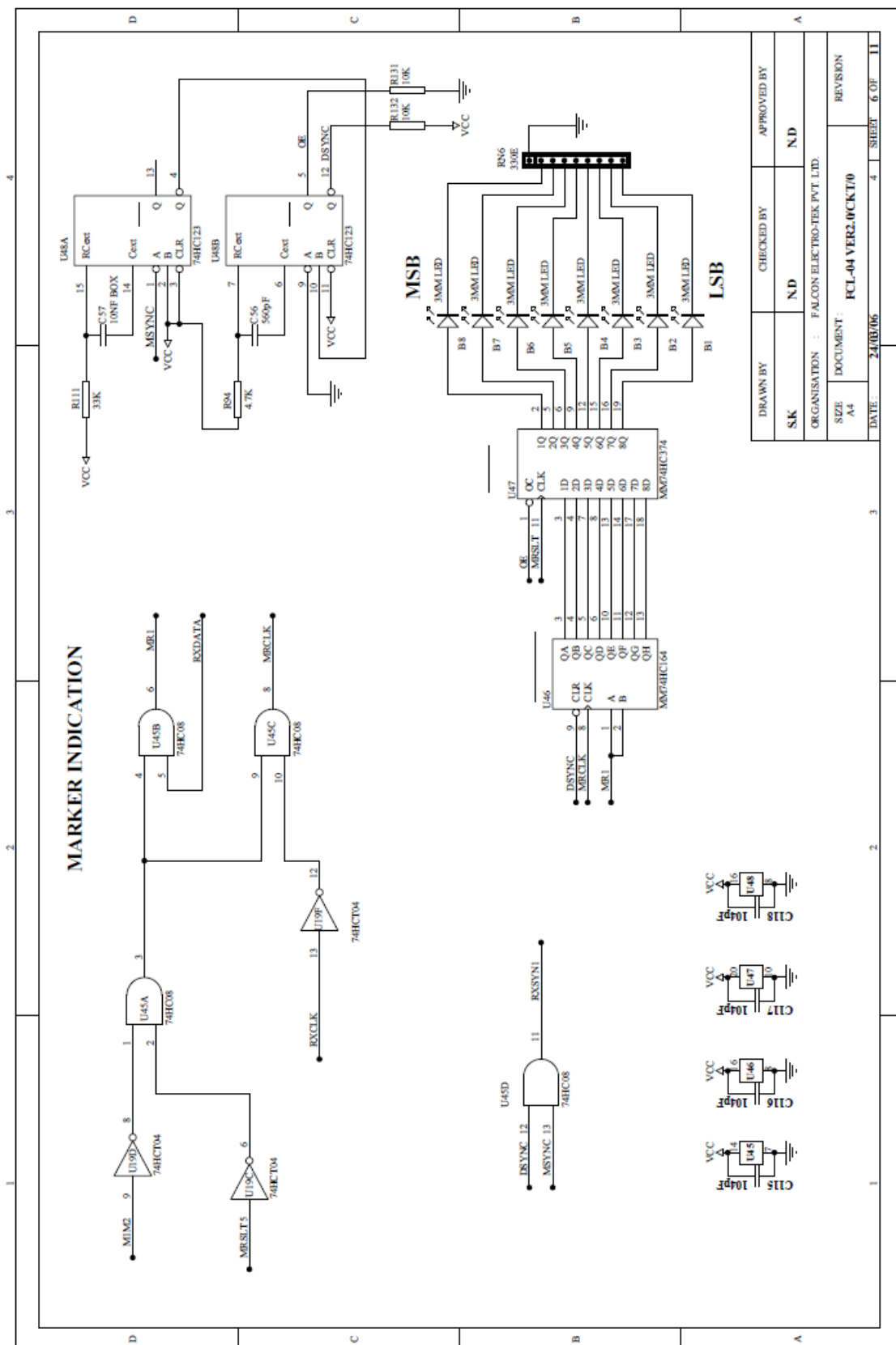
CIRCUIT DIAGRAM

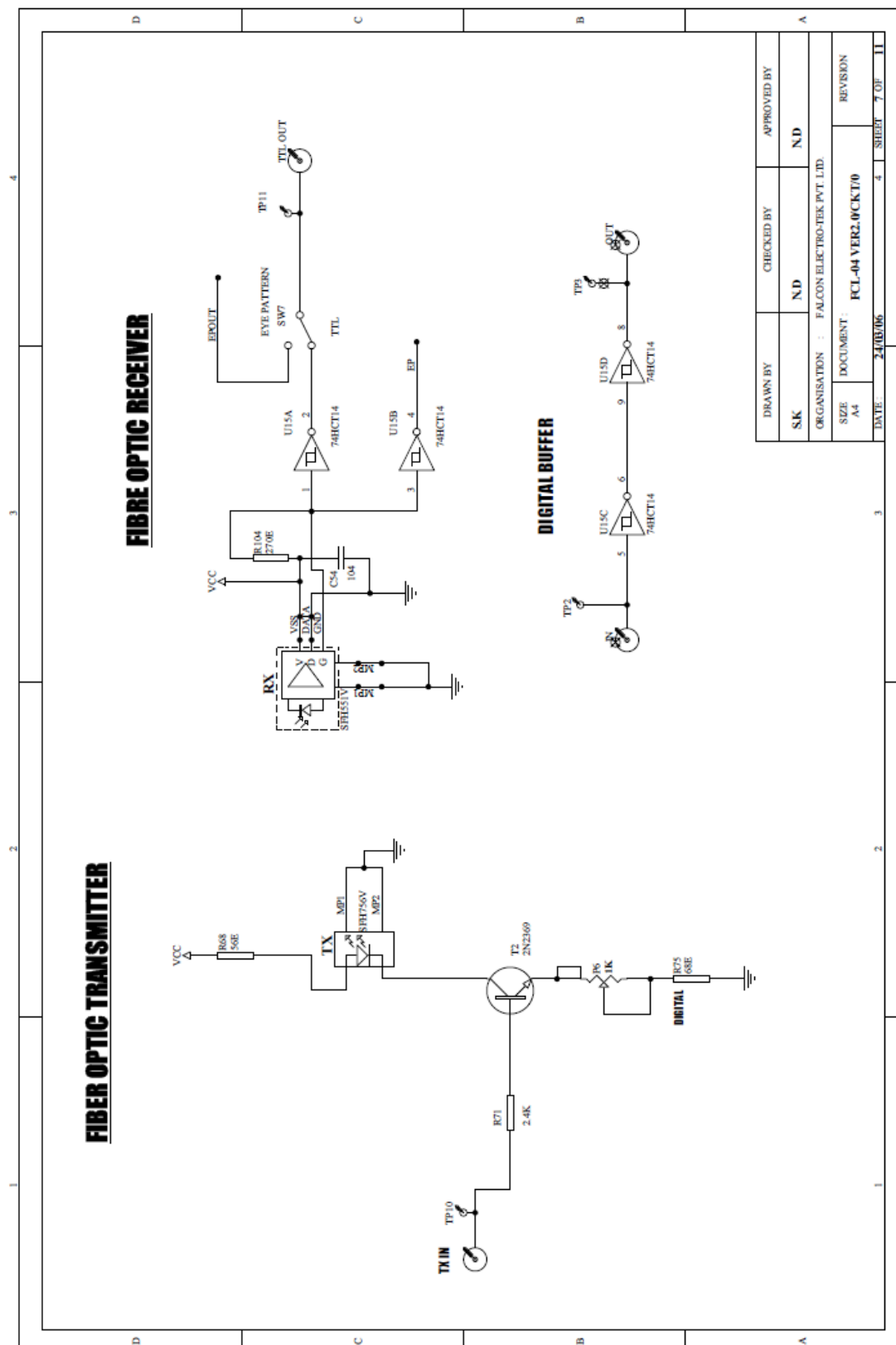


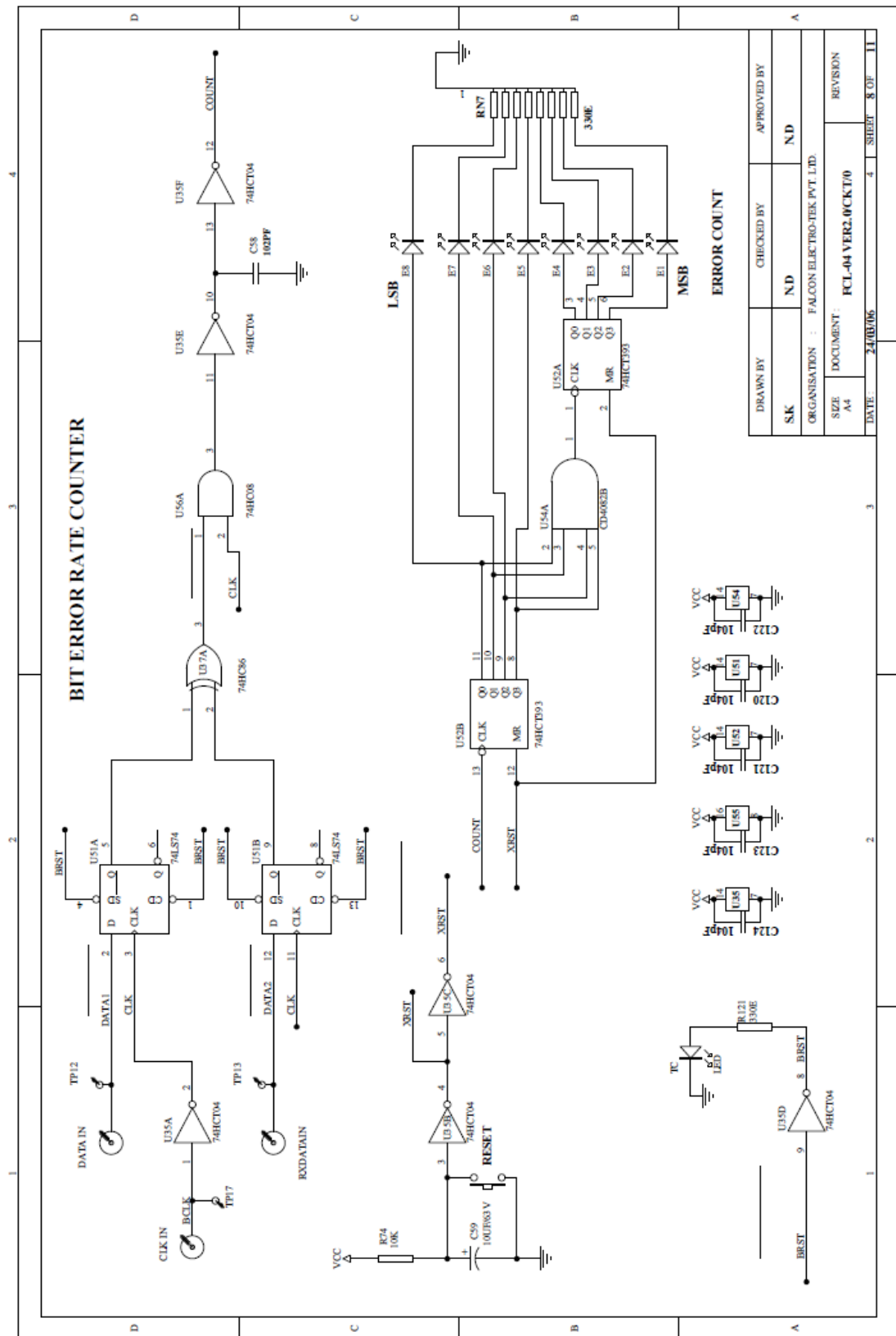


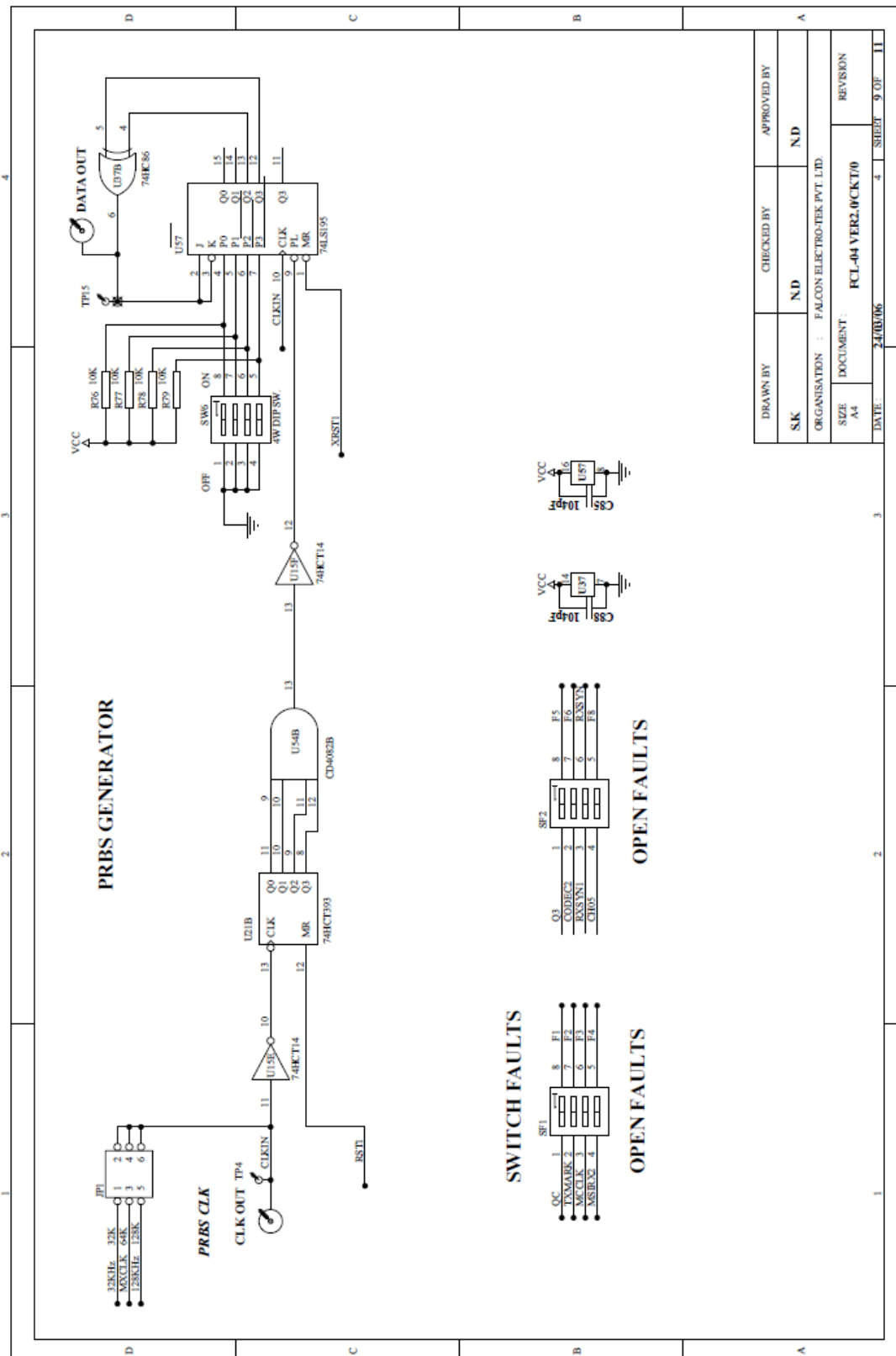


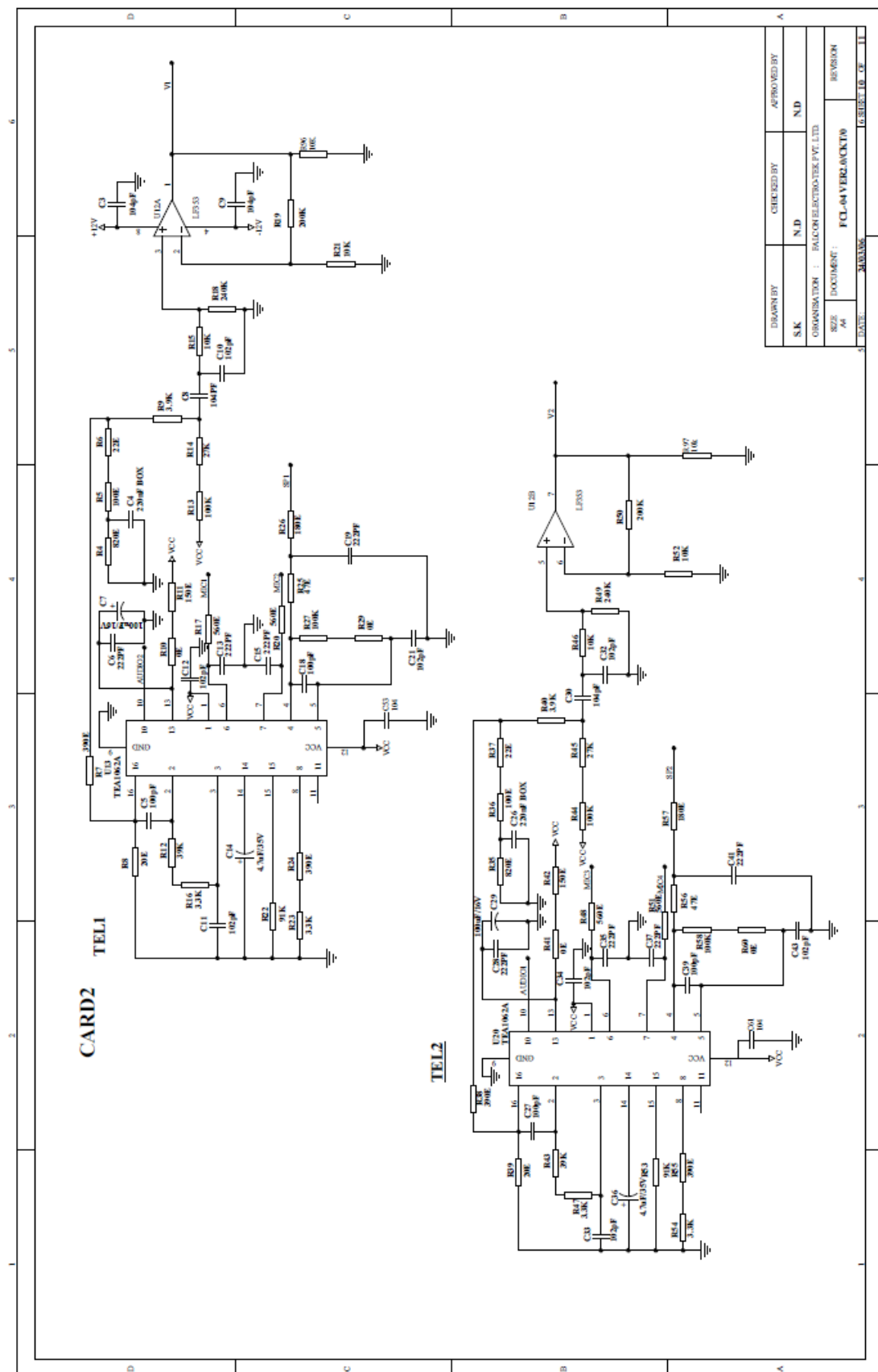


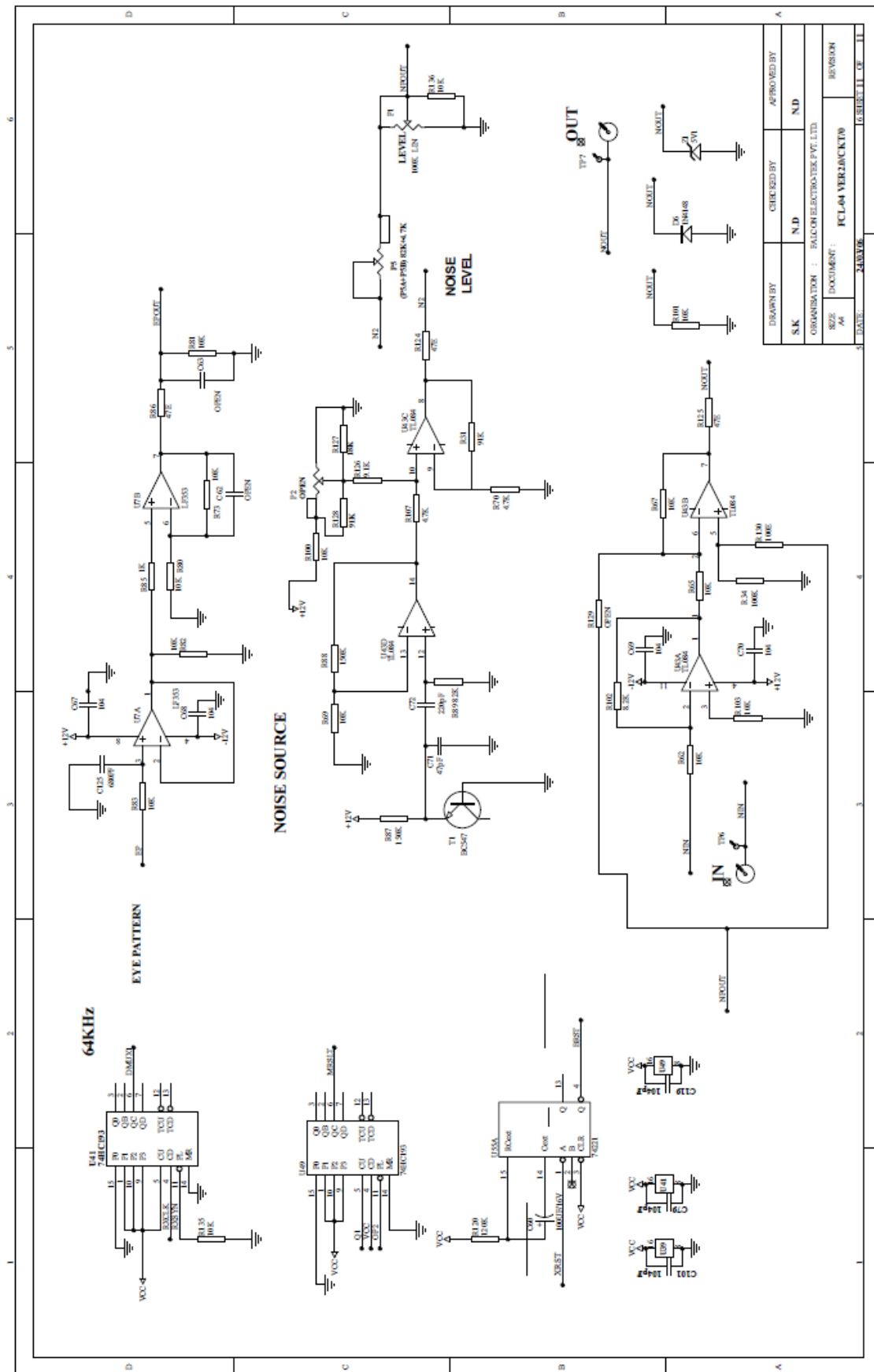












EXPERIMENT NO.1

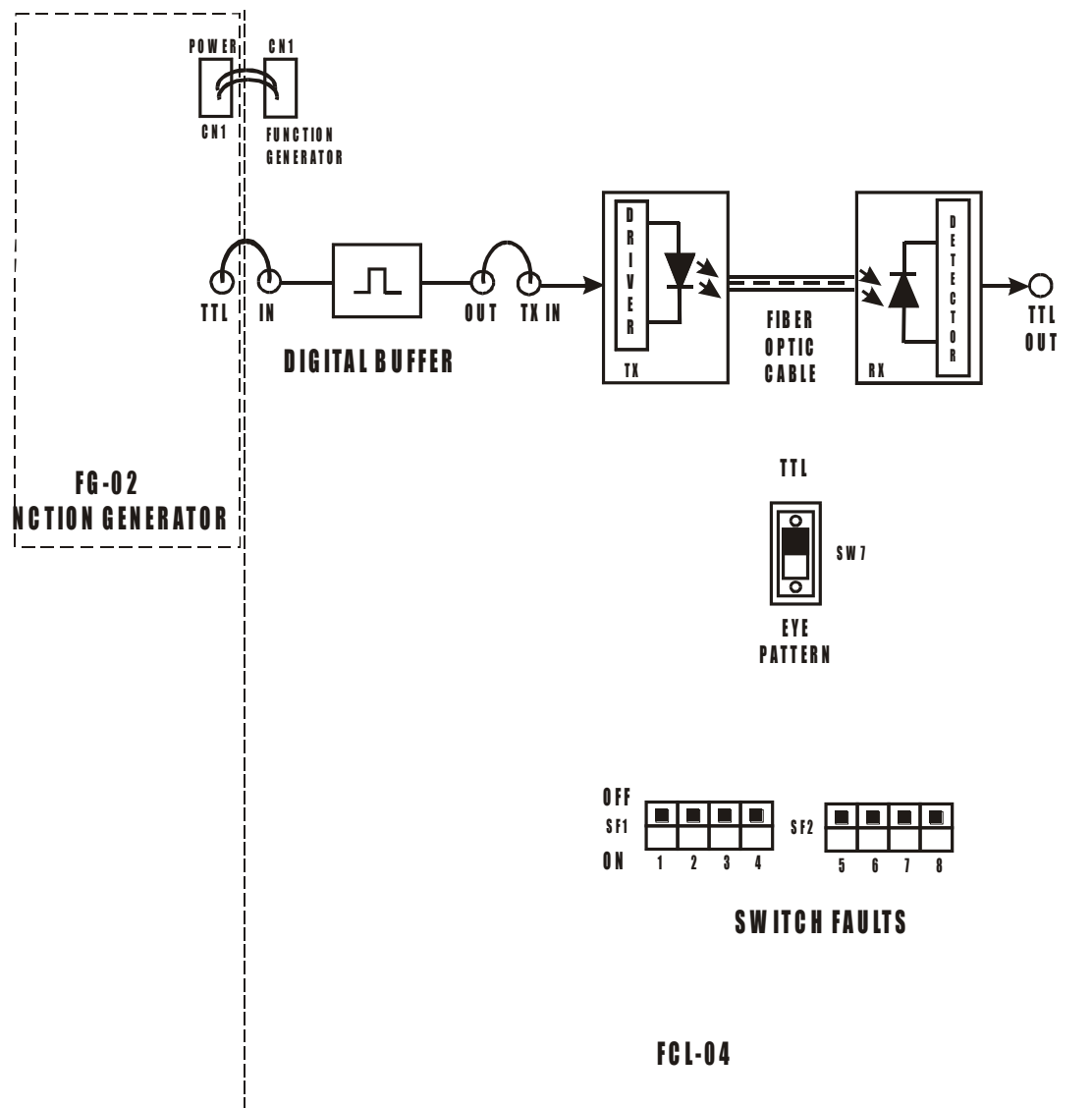


FIG. 1.1 BLOCK DIAGRAM FOR FIBER OPTIC DIGITAL LINK

EXPERIMENT NO: 1

NAME

Setting Up Fiber Optic Digital Link

OBJECTIVE

The objective of this experiment is to study 660 nm fiber optic digital link. Here you will study how digital signal can be transmitted over fiber cable and reproduce at the receiver end.

THEORY

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically, a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

Transmitter

LED, digital, DC coupled transmitters are one of the most popular variety due to their ease of fabrication. We have used a standard TTL gate to drive a NPN transistor, which modulates the LED SFH450V (950nm) source (Turns it ON and OFF).

Receiver

There are various methods to configure detectors to extract digital data. Usually detectors are of linear nature. We have used a photodetector having TTL type output. Usually it consists of PIN photodiode, transimpedance amplifier and level shifter.

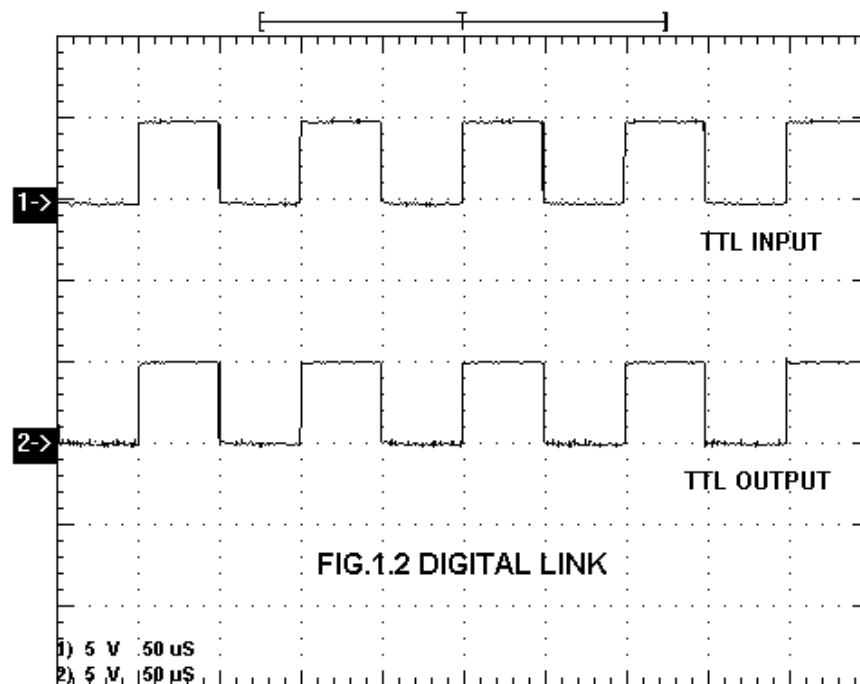
EQUIPMENTS

FCL-04
FG-02 with power cable
1 Meter Fiber cable
Patch chords
Power Supply (Use only one provided)
20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.1.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.
- Connect Function Generator FG-02 to FCL-04 using power cable.
- Turn on the power supply.
- Keep the switch **SW7** to **TTL** position on FCL-04 as shown in fig.1.1.
- Connect the **TTL** Signal from FG-02 as a constant signal to the **IN** post of Digital Buffer on FCL-04.
- Connect the output of Digital Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Observe the output signal from the detector at **TTL OUT** post on Oscilloscope you should get the reproduction of the original transmitted signal.
- To measure the digital bandwidth of the link, keep the same connections and vary the frequency of the input signal from 100Hz – 1.5MHz onwards. Observe the variation in duty cycle of the received signal for each frequency reading and determine the maximum bit rate that can be transmitted on the digital link.



EXPERIMENT NO.2

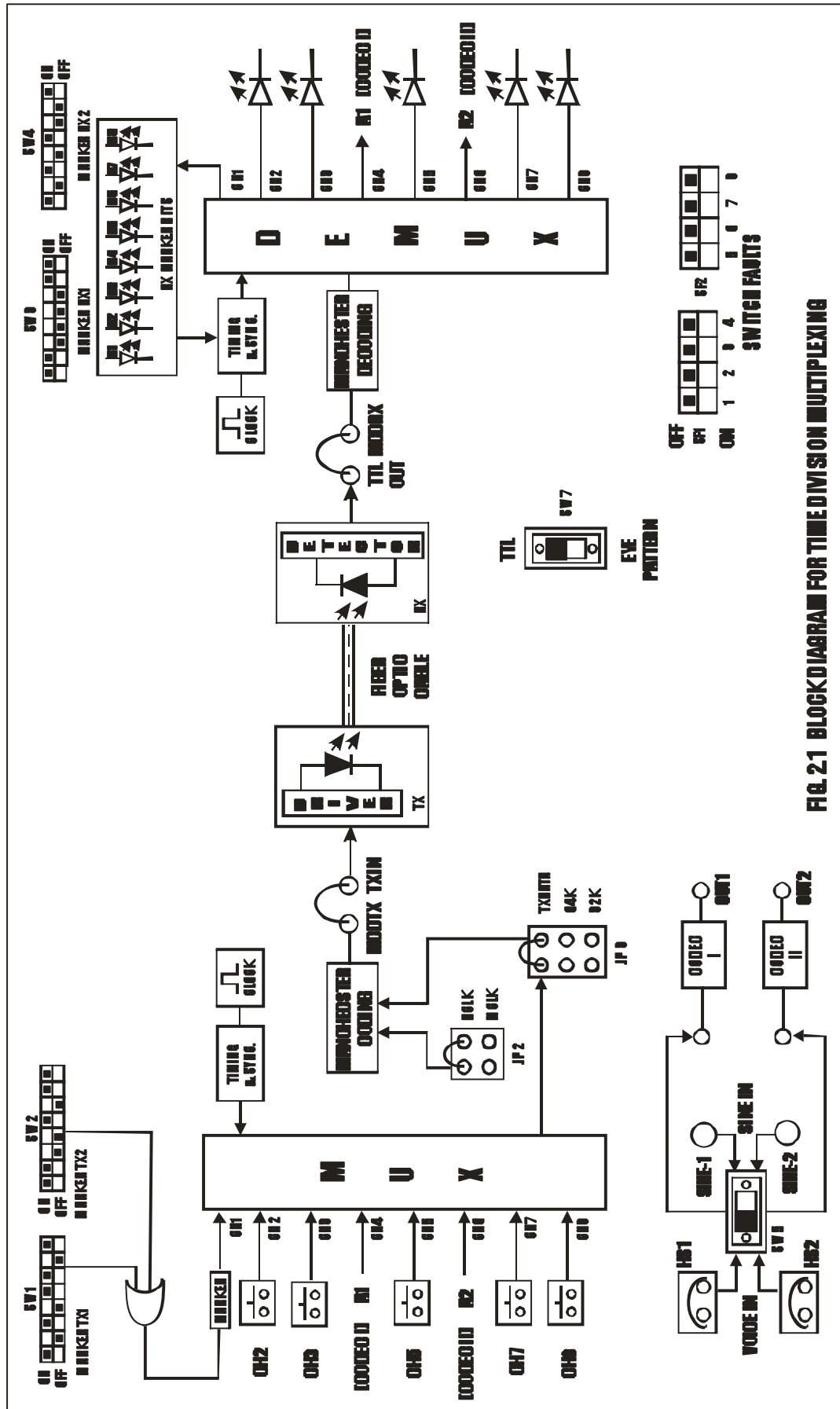


FIG. 21 BLOCK DIAGRAM FOR TIME DIVISION MULTIPLEXING

EXPERIMENT NO. 2

NAME

Study of Digital Time Division Multiplexing

OBJECTIVE

The objective of this experiment is to study simultaneous transmission of several signals using synchronous time division multiplexing.

THEORY

In case of communication systems, signals, which are transmitted usually, carry voice or video information with them & are interpreted by human eye or ears, which have slow response. Persistence of vision as well as of hearing has given rise to the concept of time division multiplexing. In time division multiplexing various signals are sampled & transmitted for a fixed duration of time one after the other. At the receiving end, these signals are extracted in the same order & form of transmission.

To implement this scheme, we have used 8 channel digital multiplexer at transmission end with clock generator for timing of signals. One channel is reserved for marker transmission; two channels for voice data transmission, five channels take their inputs from five data switches. Each channel has a data rate of 64 Kbits / Sec.

This multiplexed data is then Manchester coded & fed as digital data to the transmitter. The received digital data is first Manchester decoded & passed through a clock recovery circuit & then demultiplexed giving each signal separate in its original form & shape.

EQUIPMENTS

- FCL-04
- 1 Meter Fiber cable
- Patch chords
- Telephone Handsets
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.2.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.
- Keep the switch **SW5** to **VOICE IN** position, **SW7** to **TTL** position on FCL-04 as shown in fig.2.1.
- Keep the jumpers **JP2** & **JP3** on FCL-04 as shown in fig.2.1.
- Switch ON the power supply.
- Connect the post **MCDTX** to the **TX IN** post on FCL-04.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect detected signal **TTL OUT** to post **MCDRX**.
- Connect Telephone handsets to posts **HS1** & **HS2**.
- Set **MARKER TX1** & **MARKER TX2** each for bit pattern shown in fig.2.1 using **SW1** & **SW2** respectively.
- Set **MARKER RX1** & **MARKER RX2** each for bit pattern shown in fig.2.1 using **SW3** & **SW4** respectively.
- Observe the time division multiplex data at **TDMTX** on Oscilloscope.
- Carefully observe the time duration for which each channel is selected. Observe & measure the frame period.
- Press either of the Channel keys (CH2, CH3, CH5, CH7 & CH8) and observe how data is transmitted in the corresponding time slot. Thus, you can observe the signals at different points of the transmitter section.
- Observe the Manchester coded data at **MCDTX**. This data is transmitted through the fiber. The received data, which is still in Manchester coded form, is available at **MCDRX** & **TDMRX** signals with respect to **TDMTX**.
- Observe the data transmission by pressing keys (CH2, CH3, CH5, CH7 & CH8) & observing the corresponding LEDs lit up.
- The voice input at one mouthpiece can be heard at the earpiece of another handset. Observe this TDM effect.
- Repeat the above procedure for marker1 & 2 as given in the marker settings table below.

Note

The students can form hundreds of other combinations apart from the marker settings table. Few settings may not give proper TDM o/p. Avoid 4 consecutive 1's or 0's in a single marker setting to get proper TDM data.

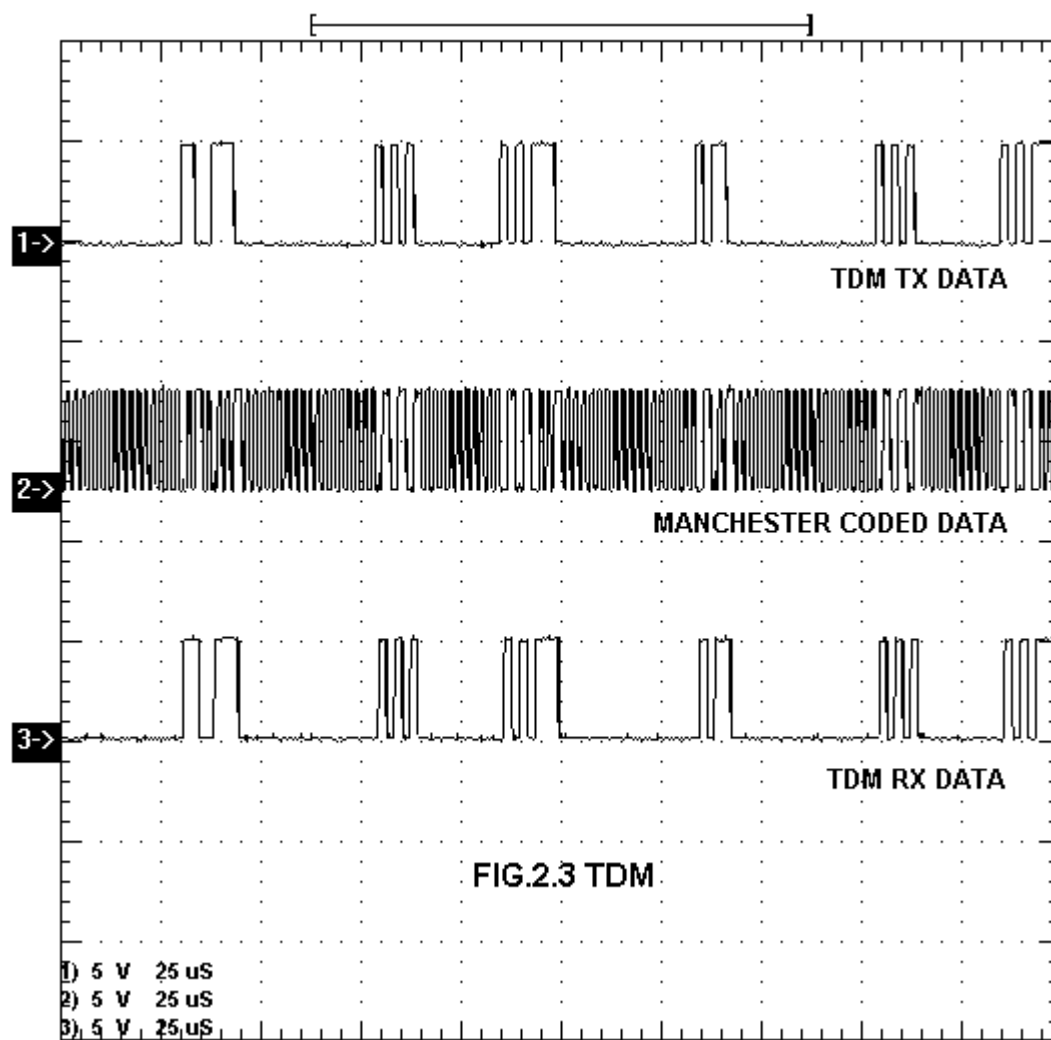
SWITCH FAULTS

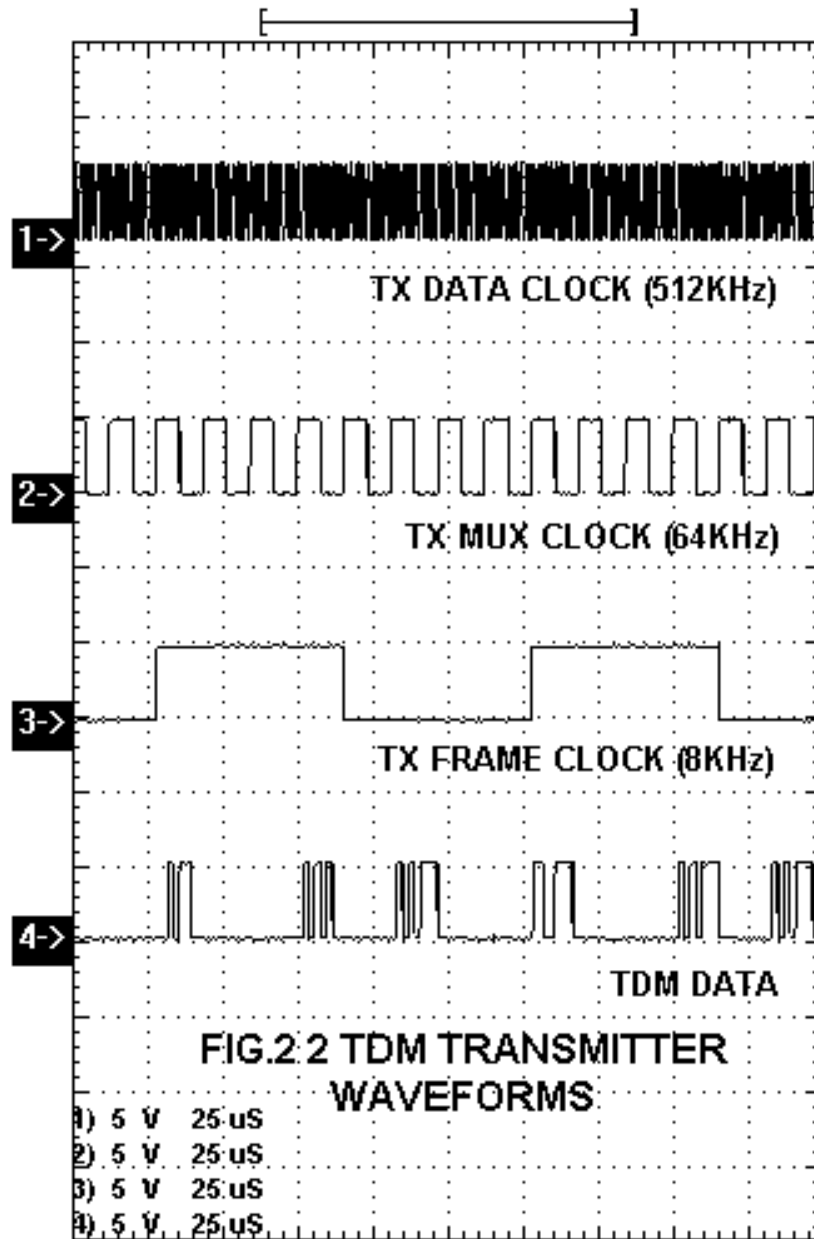
NOTE: Keep the connections as per the procedure. Now Switch ON corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1 (SF1)** in Switch Fault section to **ON** position. Clock 8KHz of U16 (4051) is removed. Output channels loose synchronization. All channels are affected.
- Put switch **2 (SF1)** in Switch Fault section to **ON** position. Transmitter markers are removed at the U16 (4051) mux I/P. Hence data is transmitted without marker & hence no sync pulse at receiver and synchronization is lost at Output.
- Put switch **3 (SF1)** in Switch Fault section to **ON** position. This will open pin 2 of U28a (XOR) in Manchester coding section. MCDTX Output gets disturbed at TP5 and hence Output channels at receiver get disturbed. (No coding takes place & hence MCD clock recovery is disturbed and no demultiplexing takes place.
- Put switch **5 (SF2)** in Switch Fault section to **ON** position. This will open pin 5 of U36 (74154) demux IC. The channel 5 is not enabled at Output. No LED indication for channel 5.
- Put switch **8 (SF2)** in Switch Fault section to **ON** position. This will open pin 21 of U36 (74154) demux IC. The demux clock Q3 is removed which affects the Output channel demultiplexing, channel indication and bit indications are disturbed.

TX MARKER -1	TX MARKER -2
10010001	10110110
11011001	10110110
11011001	10001010
01011101	10001010
10101101	10001010
10101101	11001011
10111001	11001011
10110001	01101010
10011101	01100010
10010101	01011001

MARKER SETTINGS TABLE





EXPERIMENT NO.3

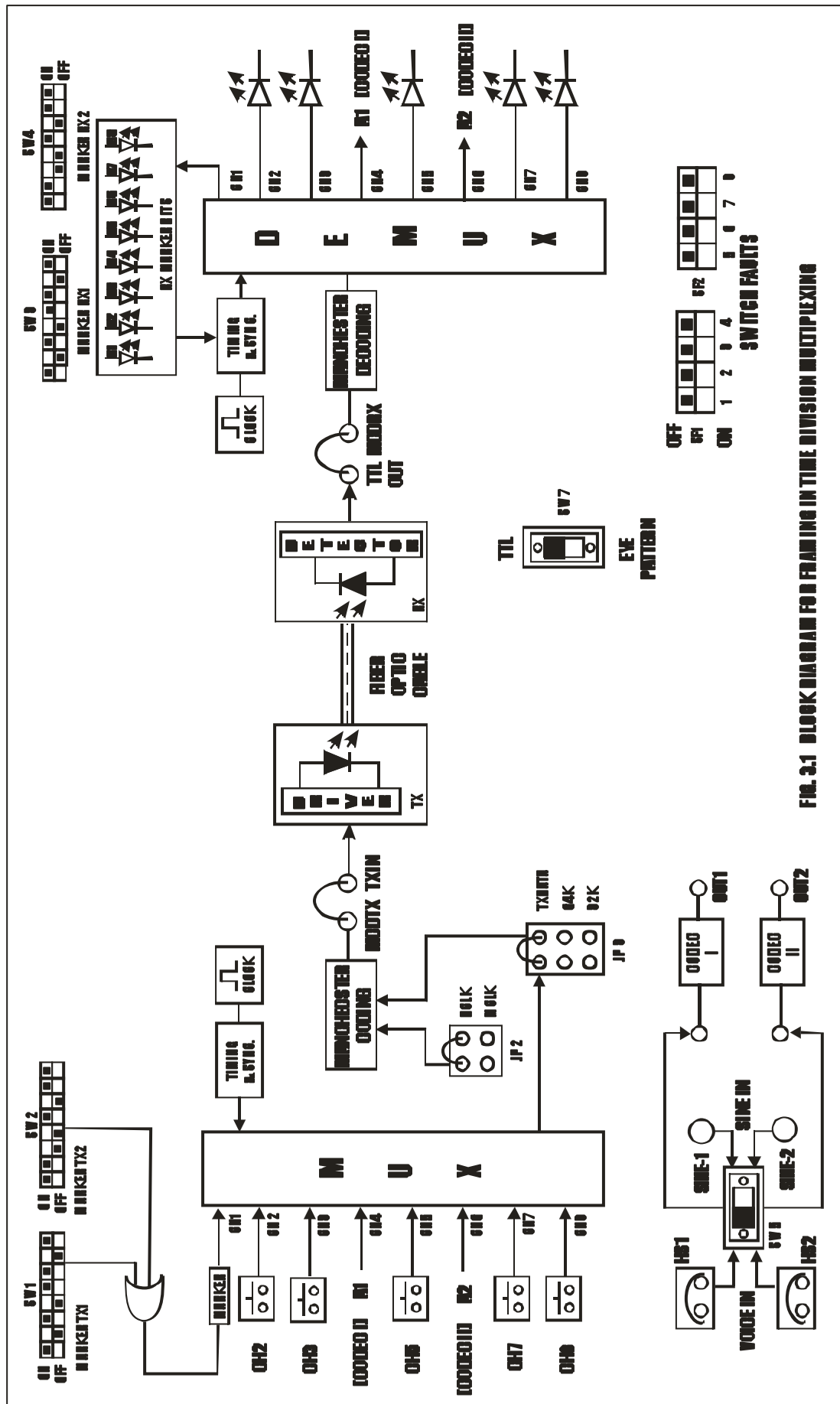


FIG. 3.1 BLOCK DIAGRAM FOR FRAMING IN TIME DIVISION MULTIPLEXING

EXPERIMENT NO.3

NAME

Study Of Framing In Time Division Multiplexing

OBJECTIVE

The objective of this experiment is to study the technique of FRAMING in time division multiplexing.

THEORY

This is an advanced experiment on Time Division Multiplexing. This experiment examines the method of synchronous multiplexing. A FRAME plays a vital role in synchronous time division multiplexing, which repeats itself after every T seconds. The frame has 'n' bits & frame rate is $1/T$ frames per sec. The total data rate is n/T bits per sec. A synchronous signal can occupy one or more bits in every frame. A signal occupying one bit per frame will have a data rate of $1/T$ bits per sec & a signal occupying 'm' bits per frame will have a data rate of m/T bits per sec.

Now the repetition rate of frame depends on channel sampling frequency. Since we are transmitting audio signals on these channels, we should sample channel at least twice the highest frequency component in audio signal, which is 4 KHz. This determines frame frequency of 8KHz. Within this period of 125 microsec, we transmit 8 channels. Hence ON period of each channel comes about $125/8$ micro sec. This corresponds to the frequency of 64 KHz. Lastly, since we transmit 8 bits of data per channel, data rate can be derived as $15.625/8=1.953125$ & $1/1.953125 = 512$ KHz.

EQUIPMENTS

- FCL-04
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

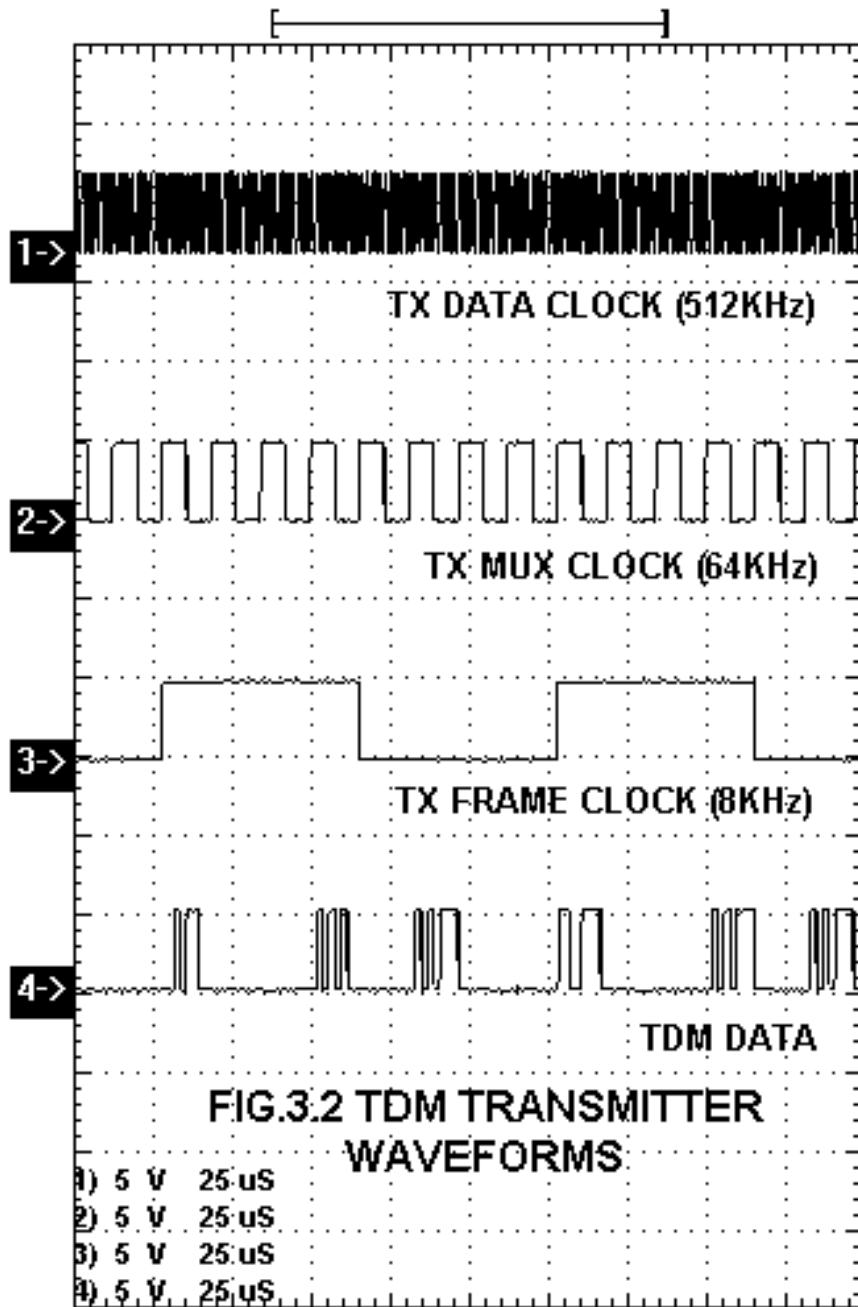
PROCEDURE

- Make connections as shown in fig.3.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.
- Keep the switch **SW5** to **VOICE IN** position, **SW7** to **TTL** position on FCL-04 as shown in fig.3.1.
- Keep the jumpers **JP2** & **JP3** on FCL-04 as shown in fig.3.1.
- Switch ON the power supply.
- Connect the post **MCDTX** to the **TX IN** post on FCL-04.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect detected signal **TTL OUT** to post **MCDRX**.
- Connect Telephone handsets to posts **HS1** & **HS2**.
- Set **MARKER TX1** & **MARKER TX2** each for bit pattern shown in fig.3.1 using **SW1** & **SW2** respectively.
- Set **MARKER RX1** & **MARKER RX2** each for bit pattern shown in fig.3.1 using **SW3** & **SW4** respectively.
- Observe Frame Frequency at **FRM CLK** and Slot frequency at **MUX CLK** simultaneously on the Oscilloscope. Draw the waveforms. You should get 8 periods of slot frequency within one period of frame frequency. Similarly observe Data Clock at **DATA CLK** and compare it with the slot frequency.
- Observe the multiplexed data at **TDMTX**. This clearly indicates the frame pattern as explained above. Compare it with the frame frequency.
- Observe the transmitted clock at **DATA CLK** and received clock at **RX DATA CLK**. The duty cycles of the two are not the same but the clocks are synchronized except for a slight transmission delay in the received clock.

SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now Switch ON corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **1 (SF1)** in Switch Fault section to ON position. Clock 8KHz of U16 (4051) is removed. Output channels loose synchronization. All channels are affected.
- Put switch **7 (SF2)** in Switch Fault section to ON position. This will open pin 1 of IC U38 (74LS125). The CODEC2 (U22) channel selection is removed and no CODEC response for analog signal or voice channel.



EXPERIMENT NO.4

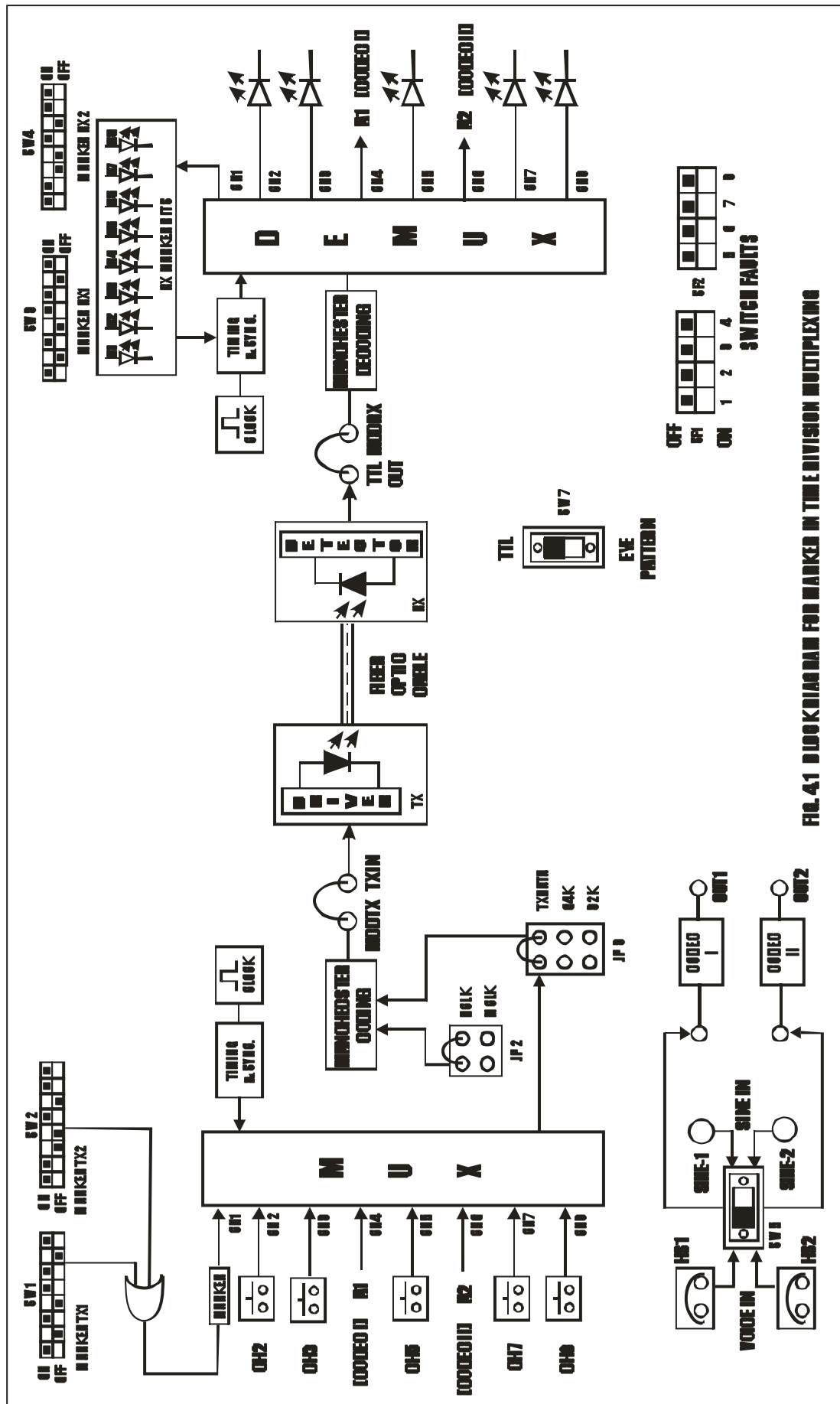


FIG. 4.1 BLOCK DIAGRAM FOR MARKER IN TIME DIVISION MULTIPLEXING

EXPERIMENT NO. 4

NAME

Study of Marker In Time Division Multiplexing

OBJECTIVE

The objective of this experiment is to study the generation & importance of Marker in Time Division Multiplexing.

THEORY

Marker used in Time Division Multiplexing is a unique bit pattern placed at some fixed position in the frame & is used to determine the start of the frame at the receiver. Sometimes a different marker is used in alternate frames to counter the possibility of data bits containing the marker sequence generating a false marker.

Double Marker- The marker is supposed to be a unique bit pattern in the data stream, which is identified at the receiver to identify the beginning of frame. Question arises, what happens if the marker pattern is contained elsewhere as data?

To avoid this situation, the receiver circuit is usually designed to detect a marker repeating itself once in a frame. It is only the repetition after a certain number of bits, which allows the bit pattern to be accepted as the marker. It is highly impossible that the data will have this pattern repeating at the same position in every frame. However, to avoid even this situation, the marker is sometimes chosen to be different in alternate frames.

EQUIPMENTS

- FCL-04
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.4.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.

- Keep the switch **SW5** to **VOICE IN** position, **SW7** to **TTL** position on FCL-04 as shown in fig.4.1.
- Keep the jumpers **JP2** & **JP3** on FCL-04 as shown in fig.4.1.
- Switch ON the power supply.
- Connect the post **MCDTX** to the **TX IN** post on FCL-04.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect detected signal **TTL OUT** to post **MCDRX**.
- Connect Telephone handsets to posts **HS1** & **HS2**.
- Set **MARKER TX1** & **MARKER TX2** each for bit pattern shown in fig.4.1 using **SW1** & **SW2** respectively.
- Set **MARKER RX1** & **MARKER RX2** each for bit pattern shown in fig.4.1 using **SW3** & **SW4** respectively.
- Observe the signal at **TDMTX** on Oscilloscope. Clearly observe the marker transmitted in the first slot of each frame. **MARKER TX1** is individually available at **MRTX1** and **MARKER TX2** at **MRTX2** post.
- Observe any of these two markers either at **MRTX1** or at **MRTX2** on Oscilloscope and change the setting of the corresponding marker and carefully observe how marker **SW1** and marker **SW2** are transmitted in alternate frames at **TDMTX**.
- Set the marker data at the receiver side using **SW3** and **SW4** similar to that at the transmitter side. **SW3** can be set to either marker **SW1** or marker **SW2**. It is the case for **SW4**.
- Observe the marker detection low going pulse at **RXMP1** for **SW3** and at **RXMP2** for **SW4**. A pulse appears in each frame period. These pulses for two markers are combined together and a single synchronization pulse is generated.
- Change the marker settings for marker1 & 2 as given in the marker setting table below.
- If the marker settings at the transmission side and the receiver side are different, you will not observe these detection pulses at **RXMP1** and **RXMP2**.
- If both the markers on transmitter and receiver side are identical, then, synchronization is achieved and one can hear clear voice from handsets.
- Now change **SW1** marker setting only at the transmitter side so that one marker instead of both will be identified at the receiver. Because of this, alternate frames will be synchronized for which **SW2** marker is detected. This will create disturbance in the audio reception.
- Now also change **SW2** marker setting at the transmitter side so that none of the markers are detected at the receiver side. This results in almost complete loss of data due to the mixing of channels.

Note

The students can form hundreds of other combinations apart from the marker settings table. Few settings may not give proper TDM o/p. Avoid 4 consecutive 1's or 0's in a single marker setting to get proper TDM data.

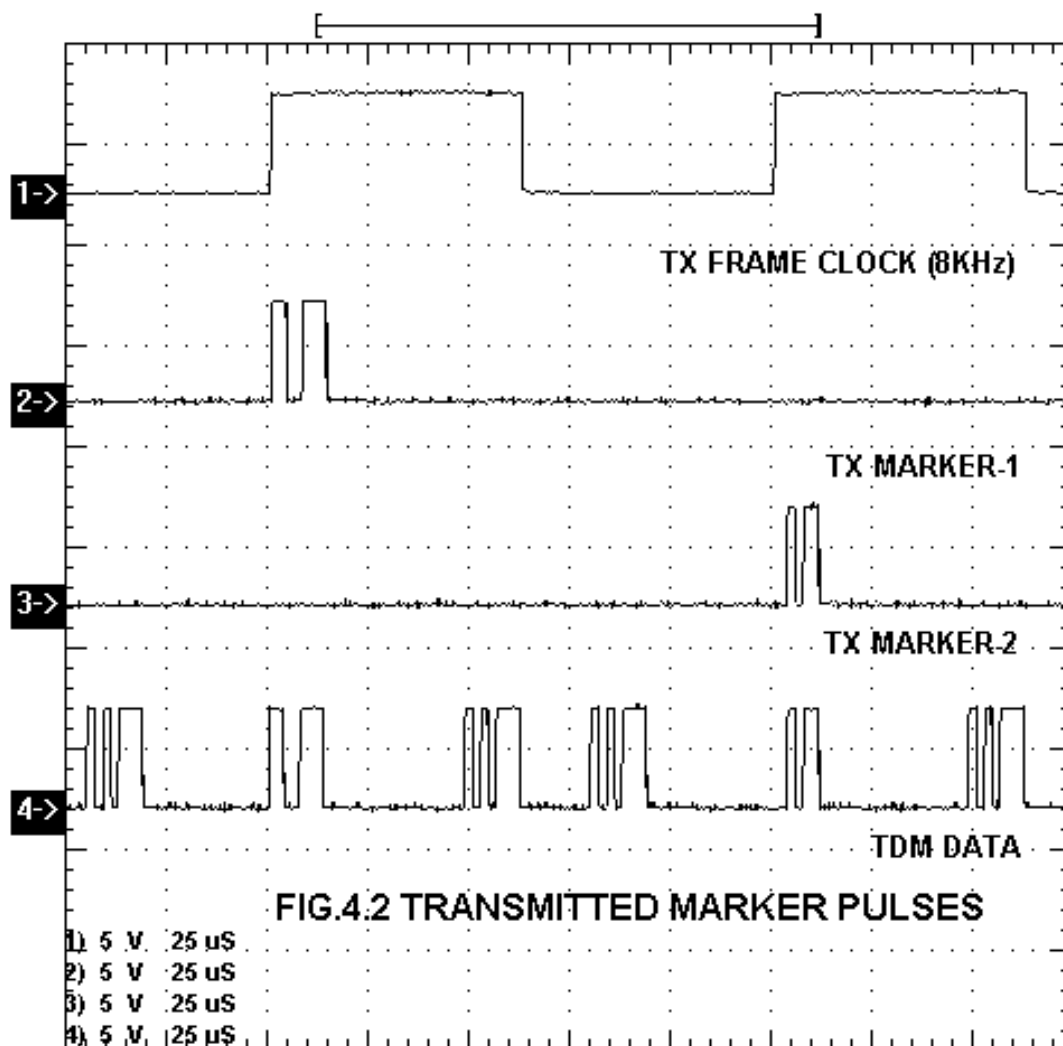
SWITCH FAULTS

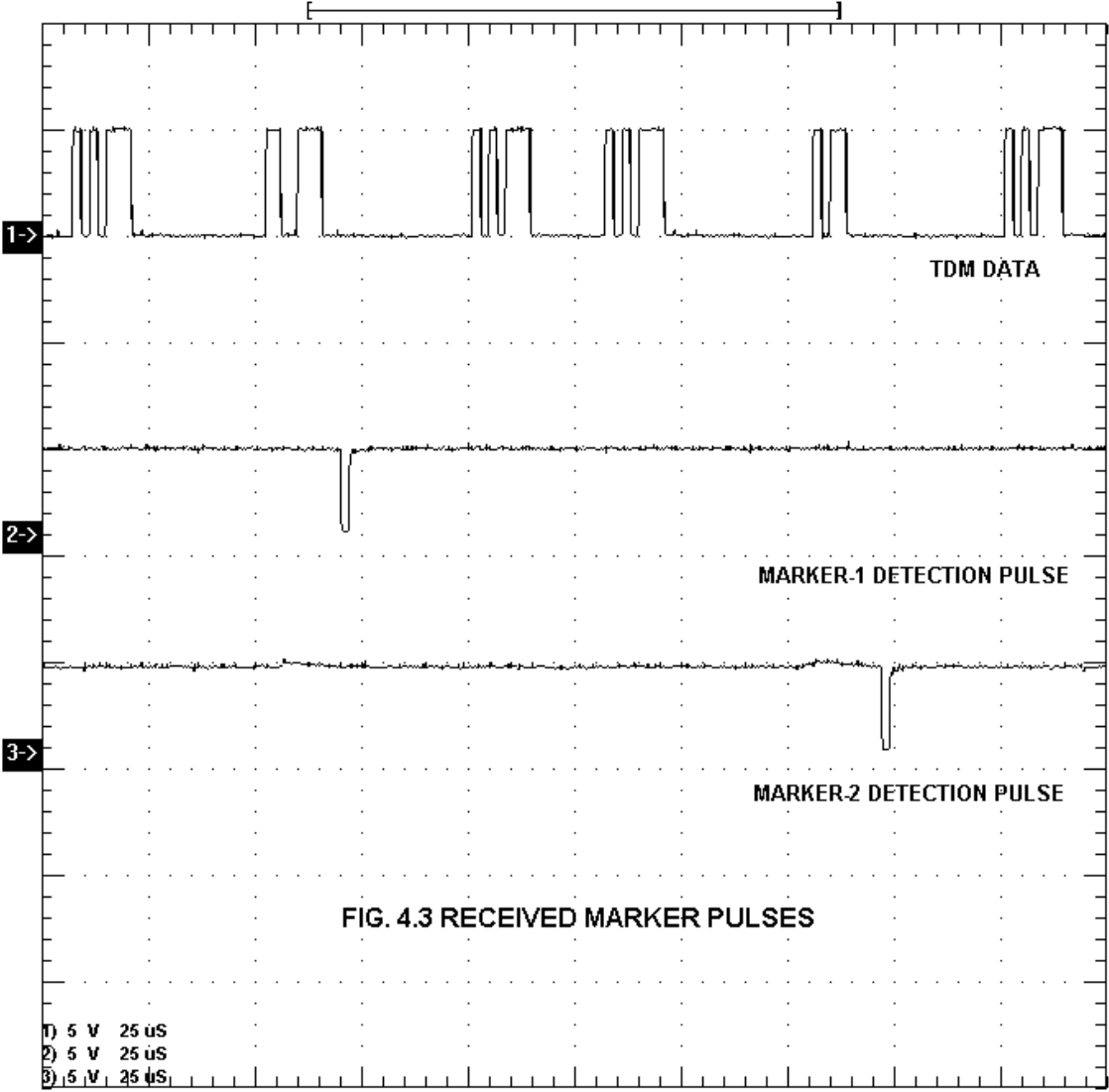
NOTE: Keep the connections as per the procedure. Now Switch ON corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **2 (SF1)** in Switch Fault section to **ON** position. Transmitter markers are removed at the U16 (4051) mux I/P. Hence data is transmitted without marker & hence no sync pulse at receiver and synchronization is lost at Output.
- Put switch **6 (SF2)** in Switch Fault section to **ON** position. This will open pin 11 of U45, RXSYN signal is removed at the receiver section & the clock recovery is disturbed, hence the receiver channels are disturbed.

TX MARKER -1	TX MARKER -2
10010001	10110110
11011001	10110110
11011001	10001010
01011101	10001010
10101101	10001010
10101101	11001011
10111001	11001011
10110001	01101010
10011101	01100010
10010101	01011001

MARKER SETTINGS TABLE





EXPERIMENT NO.5

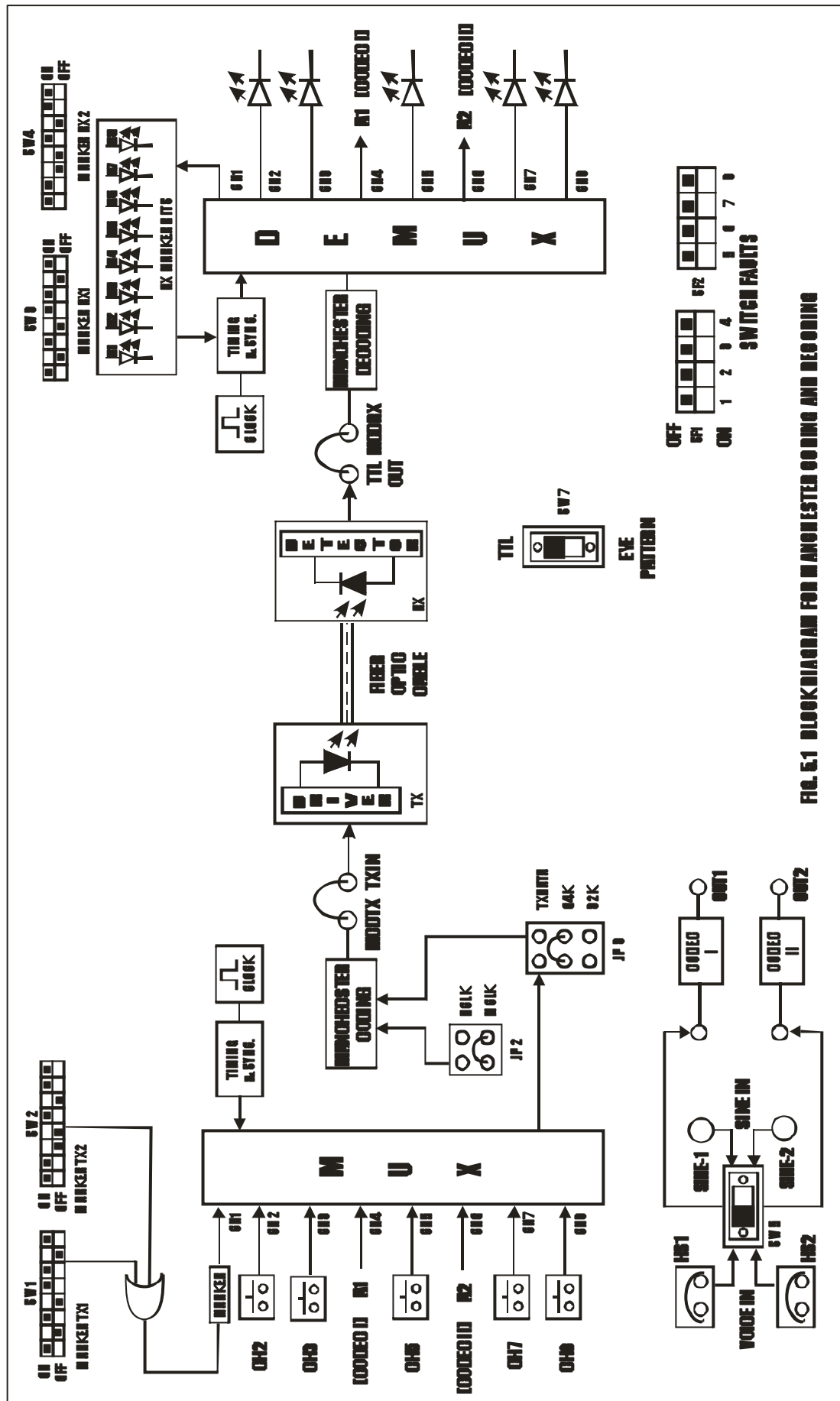


FIG. 6.1 BLOCK DIAGRAM FOR MANCHESTER CODING AND DECODING

EXPERIMENT NO.5

NAME

Study of Manchester Coding & Decoding

OBJECTIVE

The objective of this experiment is to study the techniques of Manchester Coding & decoding in digital communication.

THEORY

A fiber optic digital communication system usually consists of the following

- Optical transmitter including electrical to optical converter (E/O)
- Optical receiver including optical to electrical converter (O/E)
- Optical fiber as data transmission medium.
- Connectors, couplers & splices.
- Line coder.
- Line decoder.
- Timing recovery unit.

In digital communication, the recovery of the clock used for transmission is essential. The technique of clock recovery depends on the bit sequence of '0' to '1' & '1' to '0' transitions. Sometimes, one cannot ensure that the data to be transmitted has such transitions. To introduce the transitions in such a case, line coding is used. For low bit rate fiber optic transmission, the commonly used line coding technique is known as Manchester Coding, whereas for higher bit rates various coding schemes like mBnB coding schemes are used. Thus line coding is implemented in digital communication for the following reasons.

- For ease of timing recovery at the receiver.
- To shape the spectrum of the signal as required by various elements in the communication system.

The recovery of synchronous clock, synchronous both in terms of phase & frequency, at the receiver is a must for any synchronous digital link. A long string of zeros & ones, i.e. lack of transitions in data, could make the receiver clock lose synchronization. Line coding introduces sufficient transitions in the data for ease of clock recovery. Also a certain pattern of data, like a long string zeros or ones, would result in a strong DC component in the transmitted signal. Such a DC component is often not desirable as line coding shifts the spectrum & avoids the DC component. Several methods of coding digital transmission for unique effects are possible. Each type of coding has inherent advantages. We will consider the simplest line coding technique as Manchester Coding. The data waveform is

composed of a binary bit stream with each bit forming a cell one clock period in length: this is shown in the fig. Data & clock signals can be combined logically to form Manchester coded waveform.

The Manchester coded waveform is generated in the following manner. If the data bit is '1' the waveform will be positive for the first half of the bit cell. For a '0' data bit, the first half of the bit cell will be zero. A transition always occurs in the center of the bit cell, regardless of whether the bit is '1' or '0'.

As one can observe in the waveform for Manchester Coding in fig., higher frequency transmission is the added penalty for the reduction of DC wonder to zero & the self-clocking feature.

EQUIPMENTS

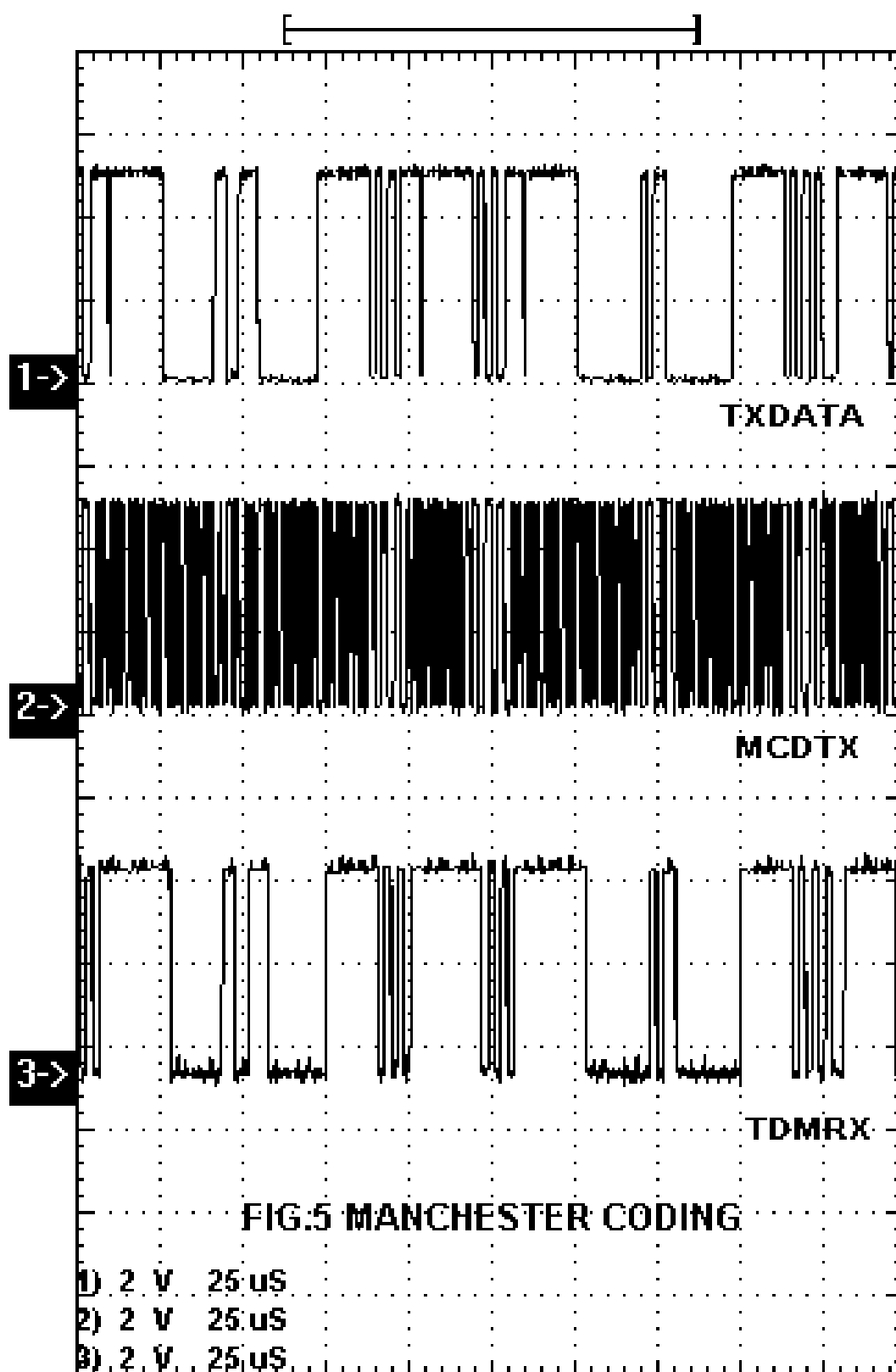
- FCL-04
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

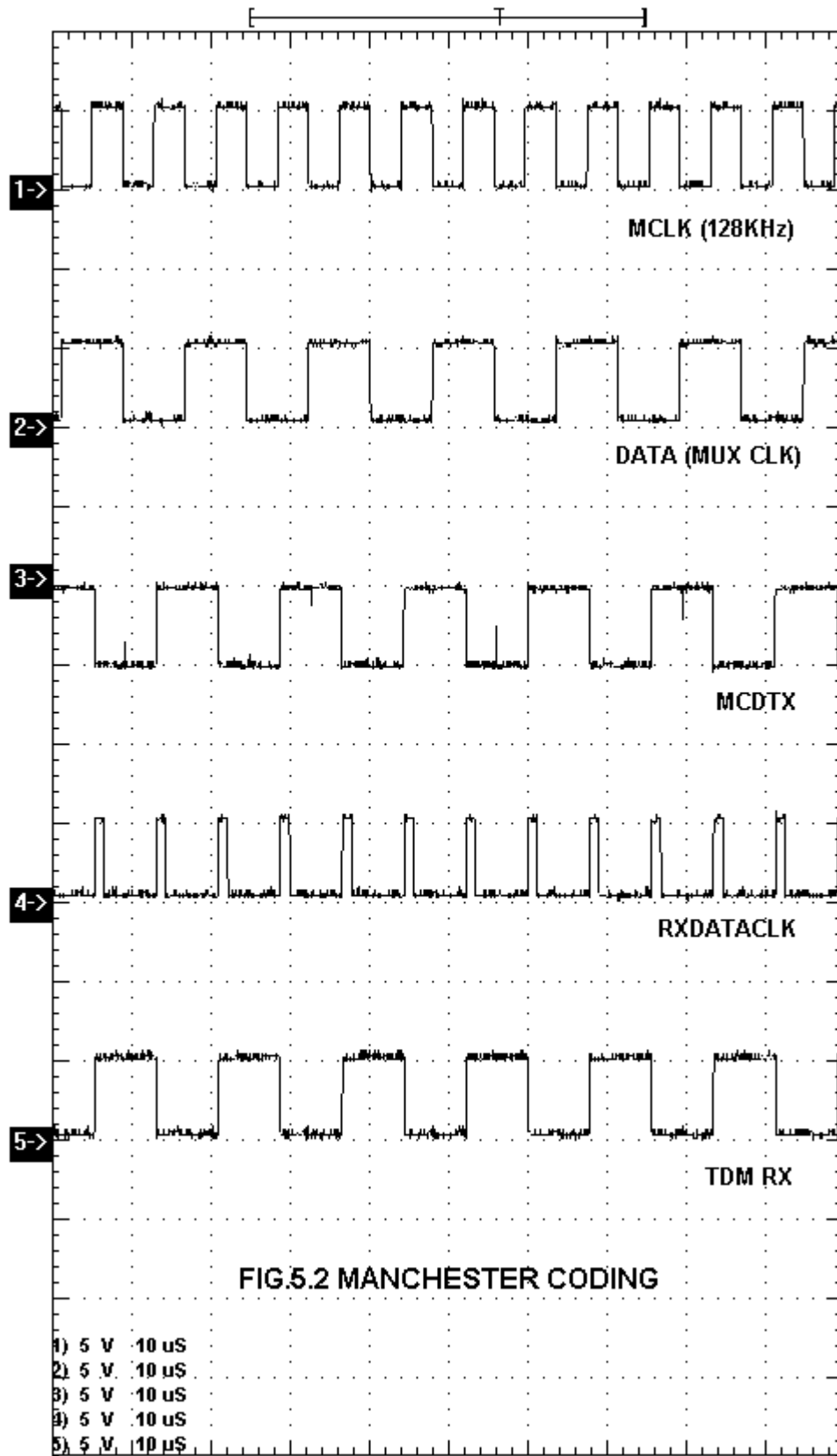
NOTE: Keep All Switch Faults In Off Position.

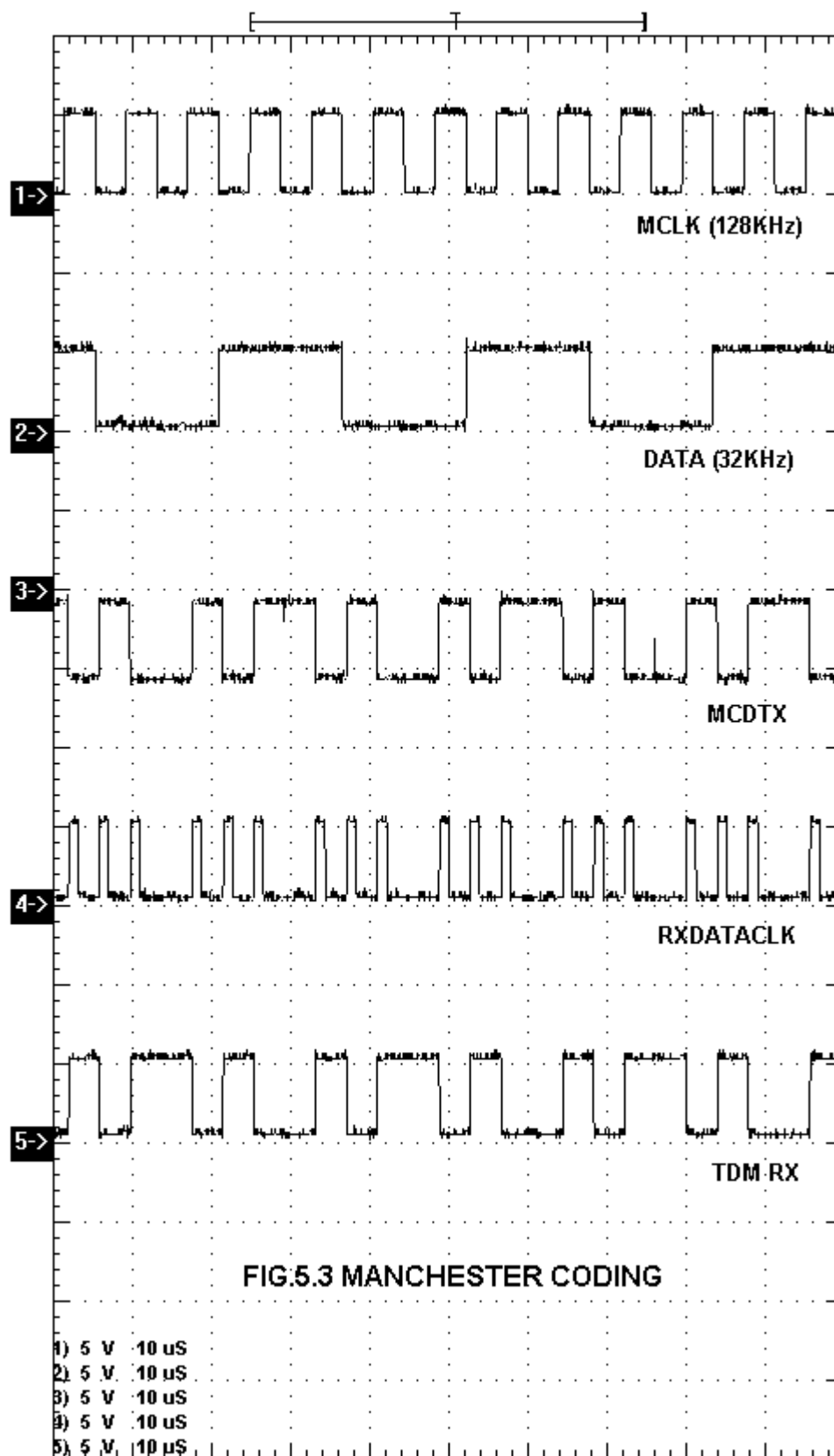
PROCEDURE

- Make connections as shown in fig.5.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.
- Keep the switch **SW5** to **VOICE IN** position, **SW7** to **TTL** position on FCL-04 as shown in fig.5.1.
- Keep the jumpers **JP2** & **JP3** on FCL-04 as shown in fig.5.1.
- Switch ON the power supply.
- Connect the post **MCDTX** to the **TX IN** post on FCL-04 through buffer.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect detected signal **TTL OUT** to post **MCDRX**.
- With current jumper setting for **JP2** and **JP3**, we have 64KHz frequency used as data input to the flip flop in Manchester coding circuit and MCLK (128KHz) frequency used as transmission clock.
- Compare signal at **DATA** post of MANCHESTER CODING section and **MCDTX post**. Signal at **MCDTX** is now Manchester coded. Also compare signal at **JP2**, which is transmission clock, and signal at **MCDTX**. Observe

- how transmissions are introduced for each clock pulse in Manchester coded data.
- Compare recovered data at **TDM RX** and original transmitted data at **TDM TX**.
 - Now, change jumper setting of **JP3** so that 32KHz frequency is now data input to the flip-flop in Manchester coding circuit. Keep the jumper setting of **JP2** unchanged so that transmission clock is still of 128KHz & Repeat the above procedure.
 - Now, change jumper settings with jumper **JP2** on **DCLK**, **JP3** on **TXDATA** & repeat the procedure with TDMTX signal as data & necessary marker settings at transmitter & receiver.
 - With above two data frequencies 64KHz & 32KHz, find out for which frequencies you get better visual effect when you compare waveforms at **TDM TX** and **MCDTX**. Find out its reason.
 - Draw waveforms at various test points in above steps.







EXPERIMENT NO.6

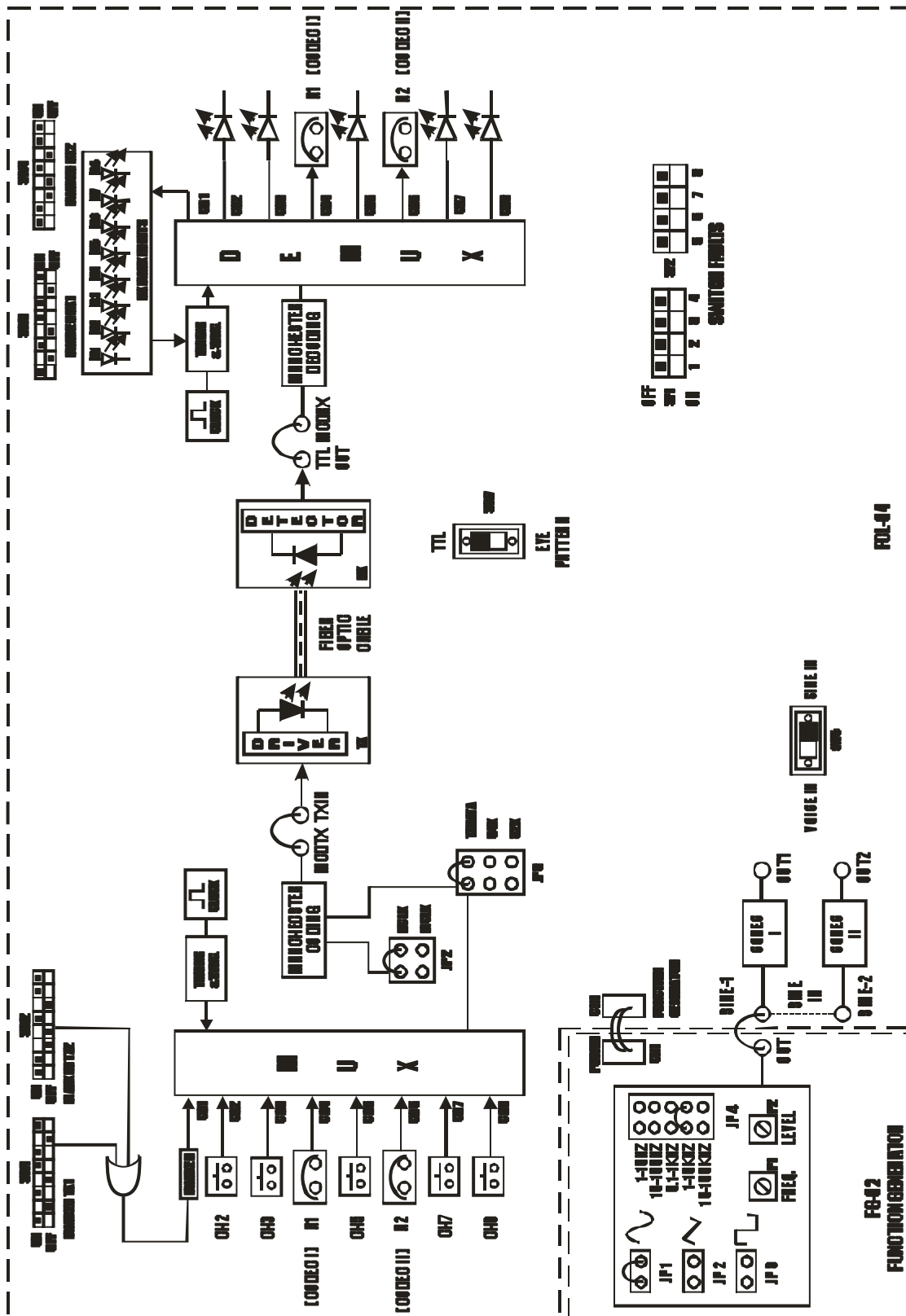


FIG. 6.1 BLOCK DIAGRAM FOR POINT-TO-POINT CODING

EXPERIMENT NO.6

NAME

Study of PCM Coding And Frequency Response Of A Codec

OBJECTIVE

The objective of this experiment is to study the linear A-law PCM coding. The analog to digital conversion as well as the reverse process and the filtering characteristics of the CODEC chip 145502 used in this kit are also studied.

THEORY

Present techniques of voice communication use standards such as A-law / μ - law companded PCM voice coding at 64 kbits/sec. When the analog speech signal is converted to pulse code modulation, it is first filtered using a low pass filter with a cut-off at about 3.4 KHz. The analog signal is sampled at 8 KHz as per the Nyquist Criteria. Each sample is quantized and coded into eight bits per sample.

Actually the voice signal features something undesirable and that is its amplitude range varies greatly from one conversation to another. If the quantization levels are uniformly spaced then it certainly creates problems. If the amplitude of the signal is small, quantization levels have to be closely spaced. This gives proper resolution. But if the signal amplitude is large then this fine resolution will result in increasing the number of code bits.

Hence, normally a technique of unequal spacings between quantization levels is used.

In the Fiber Link B kit, the audio signal from the telephone handset is processed using telephone interface chip 1062A and amplifier circuit. It is then applied to voice coder. This chip actually performs analog to digital conversion and vice-versa.

The digital data output is in pulse code modulated form. The CODEC chip used exhibits both A-law and μ -law companding techniques. Here, we have selected A-law of companding. The digital output of CODEC is connected as i/p to the multiplexer. The received digital data is again converted into analog form by the chip. The clock considerations and other features of the CODEC chip used in this kit can be studied in detail from the datasheets provided with the manual.

EQUIPMENTS

- FCL-04
- FG-02 with power cable
- 01 Meter Fiber cable
- Patch chords

- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.6.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.
- Connect the post **MCDTX** to the **TX IN** post on FCL-04.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect detected signal **TTL OUT** to post **MCDRX**.
- Set the markers such that **MARKER RX1** is set similar to **MARKER TX1** & **MARKER RX2** is set similar to **MARKER TX2** as shown in fig 6.1. Observe the time division multiplex data at **TDMTX** on Oscilloscope
- Connect Function Generator FG-02 to FCL-04 using power cable.
- Switch ON the power supply.
- Keep FG-02 jumpers **JP1** shorted; **JP2**, **JP3** open & **JP4** in **1-10KHz** position on FG-02 as shown in fig.6.1.
- Keep the switch **SW5** to **SINE IN** position on FCL-04 as shown in fig.6.1.
- Connect the **OUT** post from FG-02 as a signal to the **SINE-1** post of CODEC-I on FCL-04.
- Observe the reconstructed waveform at **OUT1** post of CODEC-I. Compare both the applied input and the reconstructed signal on Oscilloscope. Draw waveforms.
- Vary the input frequency in steps and simultaneously observe the frequency response of CODEC-I. Measure the amplitude of the output signal for each input frequency.
- Find the frequency reading after which the response of the CODEC drops. This gives the bandwidth of the CODEC.
- Repeat the same procedure for CODEC-II.
- Repeat the above procedure for marker1 & 2 as given in the marker settings table below.

Note

The students can form hundreds of other combinations apart from the marker settings table. Few settings may not give proper TDM o/p. Avoid 4 consecutive 1's or 0's in a single marker setting to get proper TDM data.

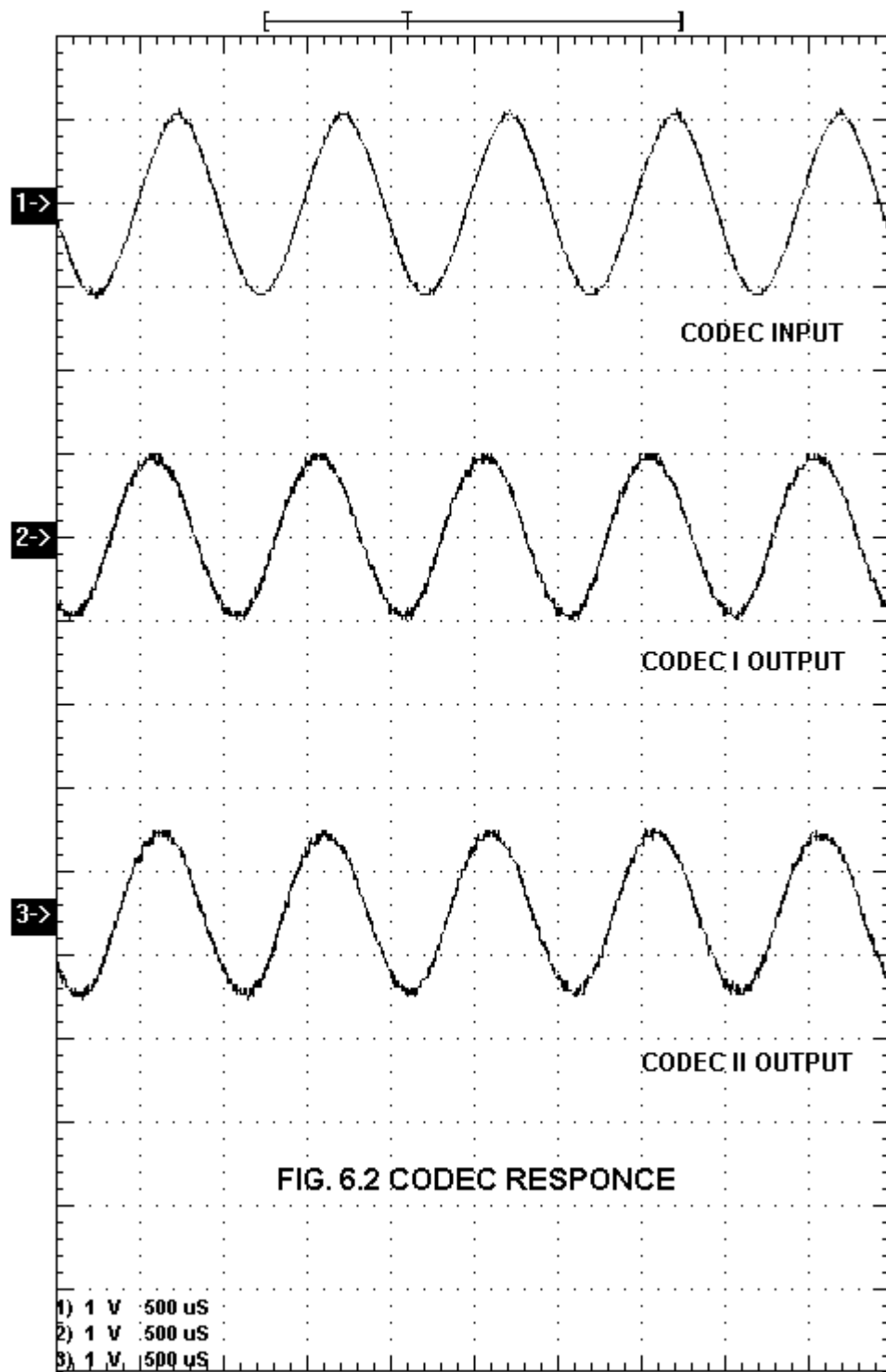
SWITCH FAULTS

NOTE: Keep the connections as per the procedure. Now Switch ON corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **4 (SF1)** in Switch Fault section to **ON** position. This will open pin 19 of CODEC2 (U22). MSIRX2 signal of CODEC2 is removed. Output sine wave of CODEC2 gets disturbed.
- Put switch **7 (SF2)** in Switch Fault section to **ON** position. This will open pin 1 of IC U38 (74LS125). The CODEC2 (U22) channel selection is removed and no CODEC response for analog signal or voice channel.

TX MARKER -1	TX MARKER -2
10010001	10110110
11011001	10110110
11011001	10001010
01011101	10001010
10101101	10001010
10101101	11001011
10111001	11001011
10110001	01101010
10011101	01100010
10010101	01011001

MARKER SETTINGS TABLE



EXPERIMENT NO.7

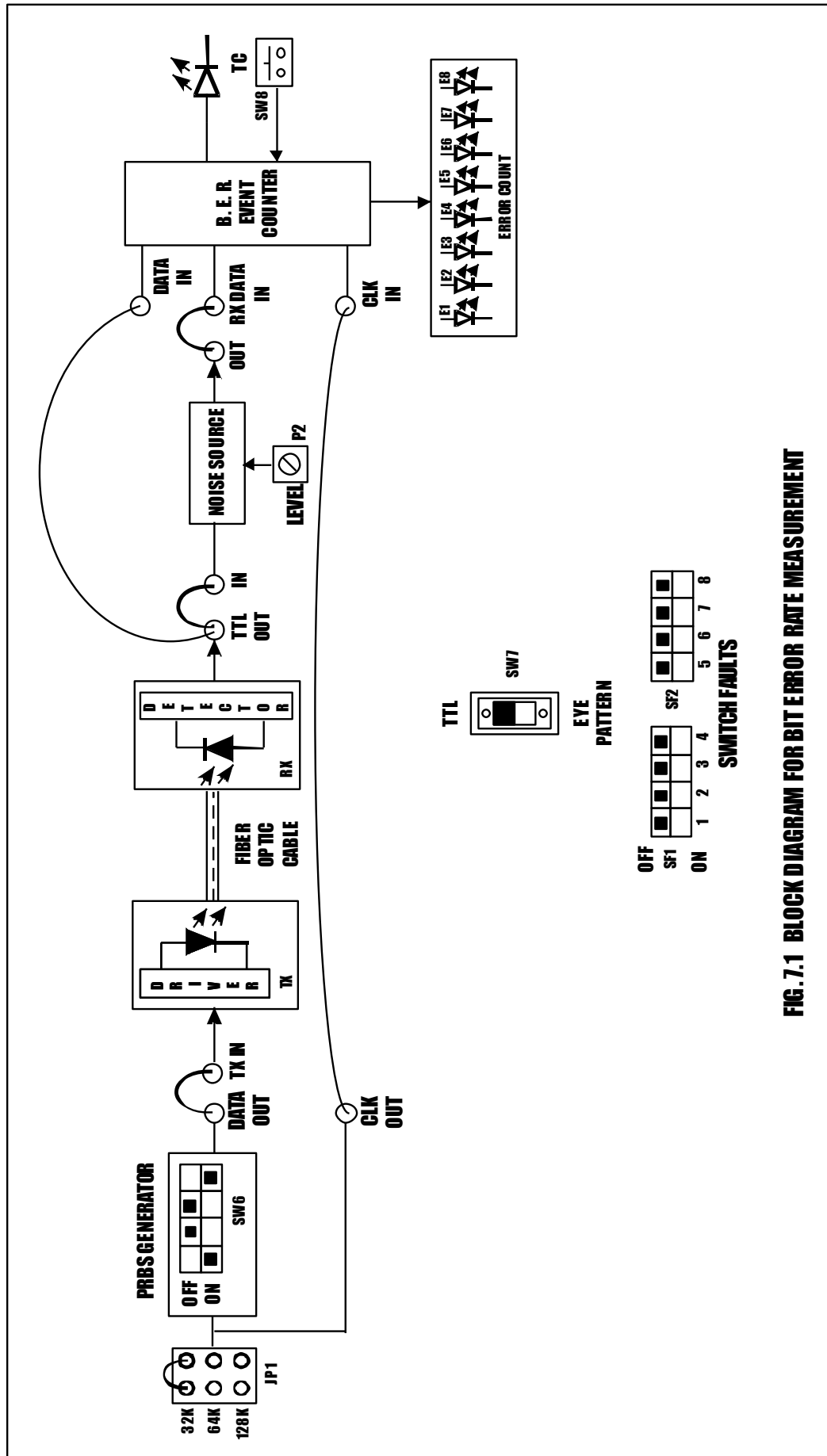


FIG. 7.1 BLOCK DIAGRAM FOR BIT ERROR RATE MEASUREMENT

EXPERIMENT NO. 7

NAME

Measurement of Bit Error Rate

OBJECTIVE

The objective of this experiment is to measure bit error rate.

THEORY

Bit Error Rate

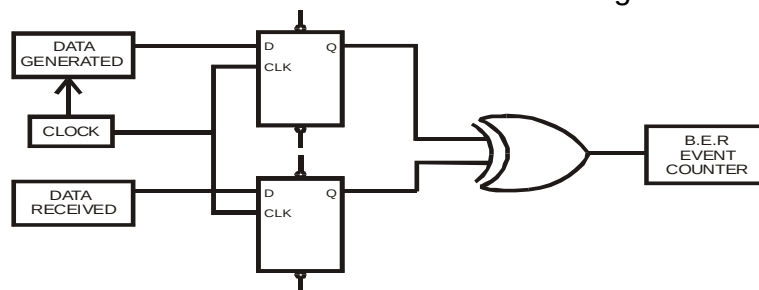
In telecommunication transmission, the bit error rate (BER) is a Ratio of bits that have errors relative to the total number of bits received in a transmission. The BER is an indication of how often a packet or other data unit has to be retransmitted because of an error. Too high a BER may indicate that a slower data rate would actually improve overall transmission time for a given amount of transmitted data since the BER might be reduced, lowering the number of packets that had to be resent.

Measuring Bit Error Rate

A BERT (bit error rate tester) is a procedure or device that measures the BER for a given transmission. The BER, or quality of the digital link, is calculated from the number of bits received in error divided by the number of bits transmitted.

$$\text{BER} = (\text{Bits in Error}) / (\text{Total bits transmitted})$$

Using a bench test setup, this is easily measured by means of a comparator in which the transmitted bits are matched in an XOR gate with the received bits. Fig shows the schematic of the device used for the following measurements.



BIT ERROR RATE TESTER (B.E.R.T.)

If the bits are alike at the XOR gate input, when clocked in from the D flip-flop, the output is low. If they are different, the XOR output goes high, causing an event count. The event counter can be set for various time periods. In general, longer the time period, more accurate the count is.

A random character generator and white noise source should be used for these measurements.

The number of bit errors is dependent upon the amount of noise entering the system. White noise, or background noise, has an average or RMS value that is

exceeded periodically by peaks that rise many times that level. These peaks exist only for a very short period of time. When the peak equals or exceeds the signal level, that is noise energy = bit energy, there is a 50/50 chance of error. The peak time periods can be calculated statistically from the error function.

In FCL-04, PRBS sequence is generated by using a 4-bit right shift register whose feedback is completed by the EX-OR gate.

Let Initially 1001 be the 4-bit switch setting on the SW6.

Clock States	D1	D2	D3	D4	
			A	B	C
1	1	0	0	1	1
2	1	1	0	0	0
3	0	1	1	0	1
4	1	0	1	1	0
5	0	1	0	1	1
6	1	0	1	0	1
7	1	1	0	1	1
8	1	1	1	0	1
9	1	1	1	1	0
10	0	1	1	1	0
11	0	0	1	1	0
12	0	0	0	1	1
13	1	0	0	0	0
14	0	1	0	0	0
15	0	0	1	0	1
16	1	0	0	1	1

Thus the sequence repeats constantly with a period corresponding to 16 clock states.

Length of sequence = $2^4 = 16$

Now the Pseudo Random Sequence pattern is C = 1010111100010011

EQUIPMENTS

- FCL-04
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.7.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.

- Keep the switch **SW7** at fiber optic receiver output to **TTL** position on FCL-04 as shown in fig.7.1.
- Keep the jumper **JP1** on FCL-04 as shown in fig.7.1.
- Switch ON the power supply.
- Keep **PRBS** switch **SW6** as shown in fig.7.1. to generate PRBS signal.
- Connect the post **DATA OUT** of PRBS Generator to the **TX IN** post.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect detected signal **TTL OUT** to **DATA IN** post of Bit Error Rate event counter & to post **IN** of Noise Source.
- Connect post **OUT** of Noise Source to post **RXDATA IN** of Bit Error Rate event counter.
- Connect post **CLK OUT** of PRBS Generator to post **CLK IN** of Bit Error Rate event counter.
- Press Switch **SW8** to start counter.
- Adjust pot **P1** for **Noise Level** to observe effect of noise level on the error count.
- Observe the Error Count LEDs for the error count in received signal in time 10 seconds.

BER MEASUREMENT

As per the definition the BER is a ratio of Errored bits (Eb) to total bits Transmitted (Tb) in a period of time t seconds.

i.e $BER = E_b / T_b$

For eg. In this experiment if PRBS data is transmitted at 32Kbits per second (i.e. jumper selection at 32KHz) for a period of 10 seconds.

So total bits transmitted in 10 seconds (Tb) = 320Kbits.

The TTL OUT data & data with noise is fed to BER counter which compares the two data inputs at each clock input.

The counter displays the Error count (Eb) on LED in 8-bit binary form (e.g 00001010), which has to be converted in decimal form (it becomes 10).

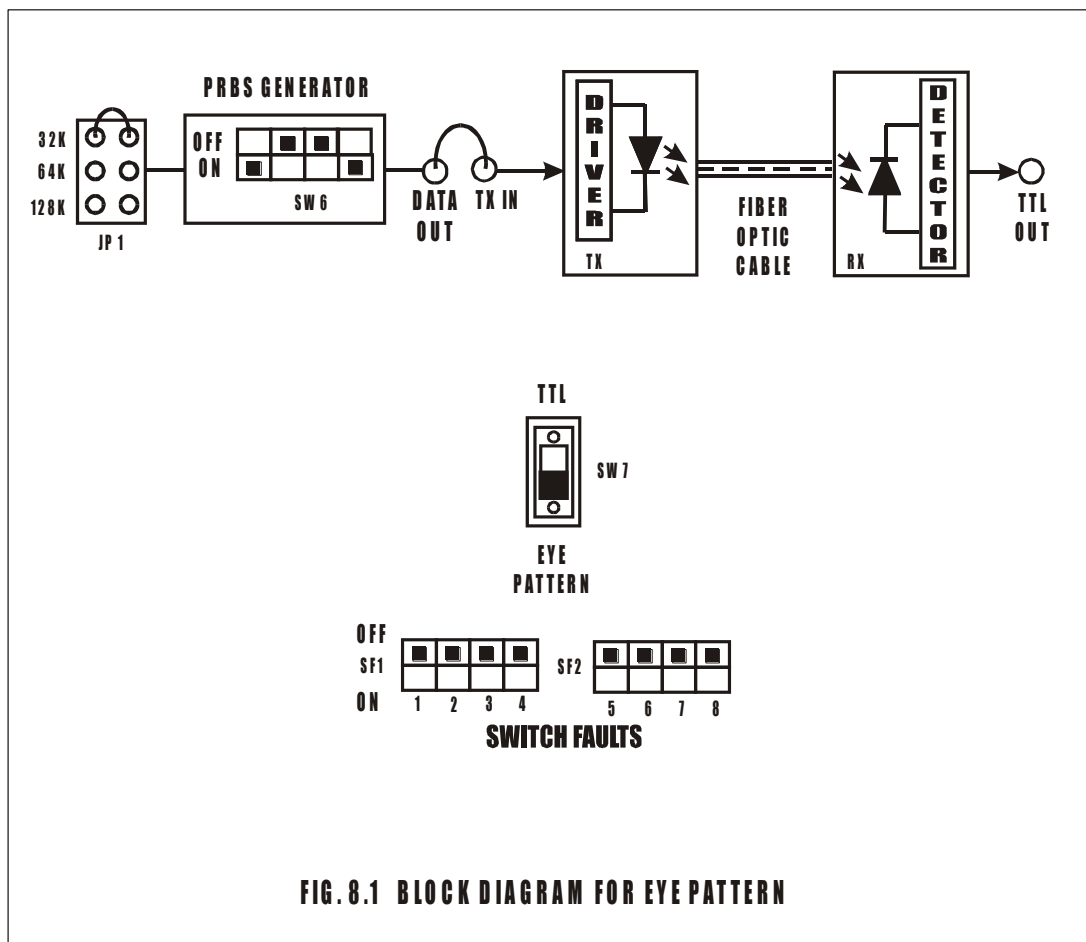
So the BER ratio then becomes

$$BER = 10 / (320 \times 10^3).$$

$$= 0.00003125$$

i.e. the channel Bit Error Rate ratio is 3.1×10^{-5} (3 / 100000) or in other words we can say that out of 100000 bits transmitted through the channel the channel gives 3 bits in error .

EXPERIMENT NO.8



EXPERIMENT NO. 8

NAME

Study of Eye Pattern

OBJECTIVE

The objective of this experiment is to study eye pattern.

THEORY

The eye-pattern technique is a simple but powerful measurement method for assessing the data-handling ability of a digital transmission system. This method has been used extensively for evaluating the performance of wire systems and can also be applied to optical fiber data links. The eye-pattern measurements are made in the time domain and allow the effects of waveform distortion to be shown immediately on an oscilloscope.

An eye-pattern can be observed with the basic equipment shown in Fig. 8-1. The output from a pseudorandom data pattern generator is applied to the vertical input of an oscilloscope and the data rate is used to trigger the horizontal sweep. This results in the type of pattern shown in Fig. 8-2, which is called the **eye pattern** because the display shape resembles a human eye. To see how the display pattern is formed, consider the eight possible 4-bit-long NRZ combinations. When these sixteen combinations are superimposed simultaneously, an eye pattern as shown in Fig. 8-2 is formed.

To measure system performance with the eye-pattern method, a variety of word patterns should be provided. A convenient approach is to generate a random data signal, because this is the characteristic of data streams found in practice. This type of signal generates ones and zeros at a uniform rate but in a random manner. A variety of pseudorandom pattern generators are available for this purpose. The word *pseudorandom* means that the generated combination or sequence of ones and zeros will eventually repeat but that it is sufficiently random for test purposes. A pseudorandom bit sequence comprises four different 2-bit-long combinations, eight different 3-bit-long combinations, sixteen different 4-bit-long combinations and so on (that is, sequences of different N-bit-long combinations) up to a limit set by the instrument. After this limit has been generated, the data sequence will repeat.

A great deal of system performance information can be deduced from the eye-pattern display. To interpret the eye pattern, follow the procedure ahead.

EQUIPMENTS

- FCL-04
- 1 Meter Fiber cable
- Patch chords
- Power Supply (Use only one provided)
- 20 MHz Dual Channel Oscilloscope

NOTE: Keep All Switch Faults In Off Position.

PROCEDURE

- Make connections as shown in fig.8.1. Connect the power supply cables with proper polarity to FCL-04 Kit. While connecting this, ensure that the power supply is OFF.
- Keep the switch **SW7** to **EYE PATTERN** position on FCL-04 as shown in fig.8.1.
- Keep the jumper **JP1** on FCL-04 as shown in fig.8.1.
- Switch ON the power supply.
- Keep switch **SW6** as shown in fig.8.1 to generate PRBS signal.
- Connect the post **DATA OUT** of PRBS Generator to the **TX IN** post on FCL-04.
- Slightly unscrew the cap of LED SFH450V (950nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect **CLK OUT** of PRBS Generator to **EXT. TRIG.** of oscilloscope.
- Connect detected signal **TTL OUT** to vertical channel **Y** input of oscilloscope. Then observe EYE PATTERN by selecting EXT TRIG KNOB on CRO as shown in fig 8.2. Observe the Eye pattern for different clock frequencies. As clock frequency increases the EYE opening becomes smaller.

